

Chapter Three

Transformers

Introduction

- The transformer transfers electrical energy from one circuit to another through the medium of a pulsating magnetic field that links both circuits.
- The ac power systems are widely used principally because of the transformer.

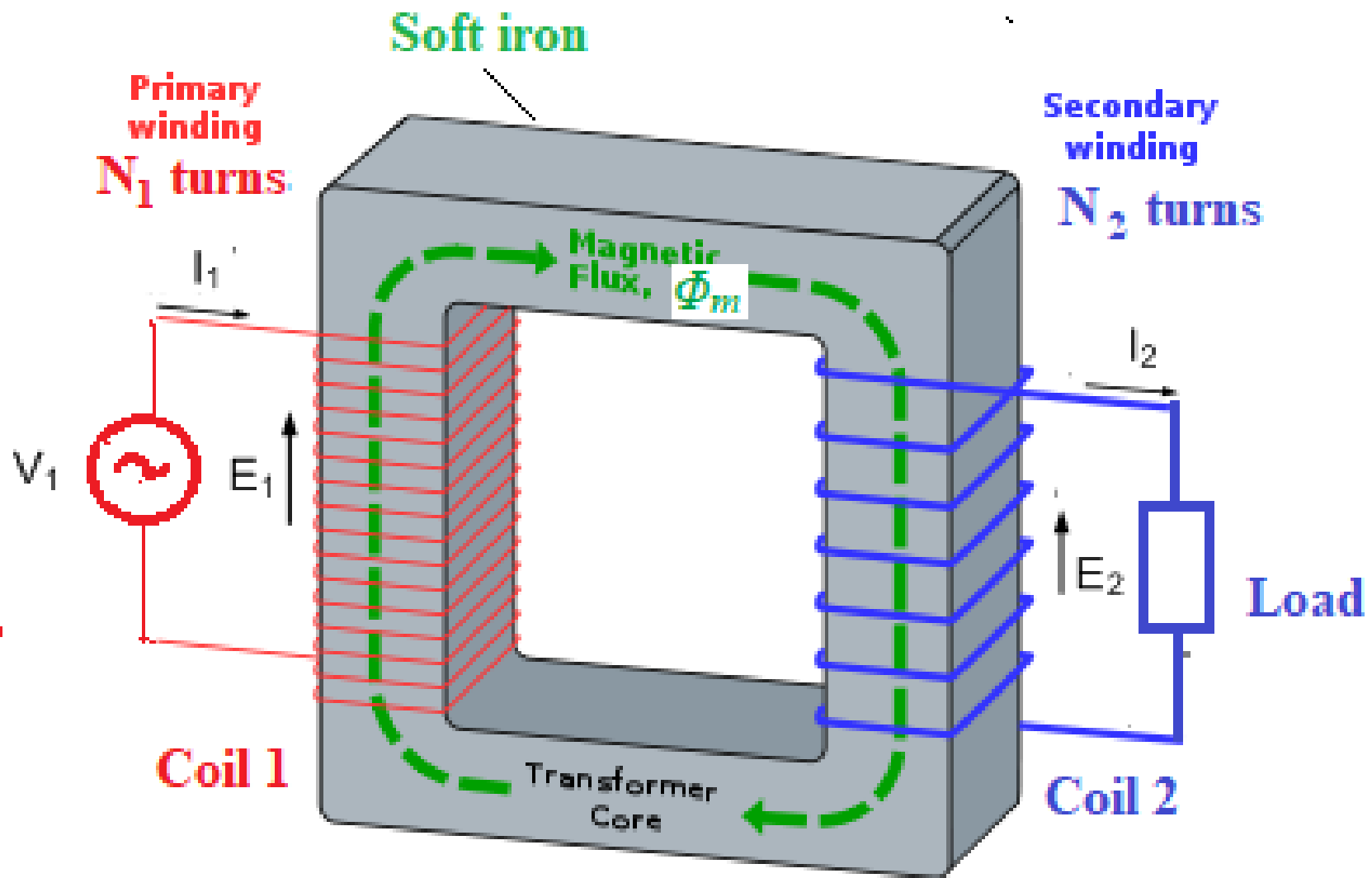
- It enables us to achieve the following in power systems:
 - to produce power at economical voltages
 - to transmit power at economical voltages and
 - to distribute power safely in factories and homes.

- Transformers are also used in low-power, low-current electronic and control circuits
 - to provide impedance matching between a source and its load for maximum power transfer
 - to isolate one circuit from another

- to isolate direct current while maintaining ac continuity between two circuits and
- to provide reduced ac voltages and currents for protection, metering, instrumentation and control.

Principle of operation of a transformer

- The transformer is a straightforward application of Faraday's law of electromagnetic induction.
- General arrangement of a 1-phase transformer is shown below.



General arrangement of 1-phase transformer

- An alternating voltage applied to coil 1, causes an alternating current to flow in the coil and this current produces an alternating flux in the iron core.
- A portion of the total flux links the second coil.
- The alternating flux induces a voltage in the second coil.

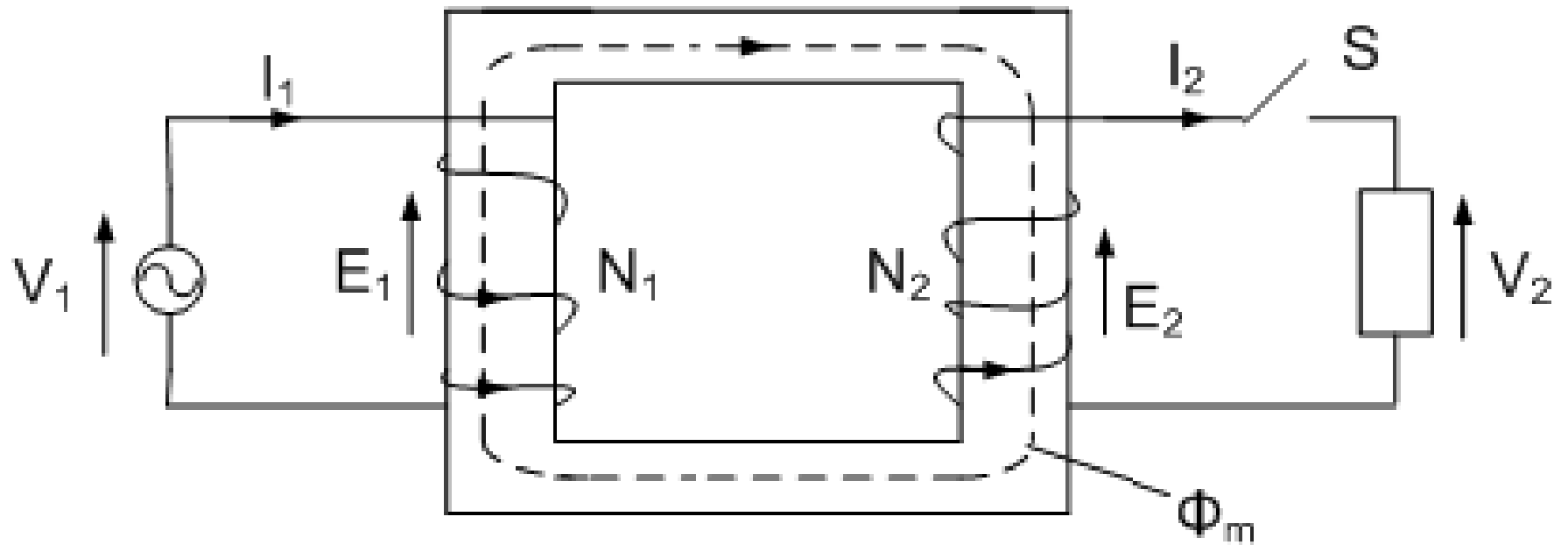
- If a load should be connected to the coil, this voltage would drive a current through it.
- Energy would then be transferred through the medium of magnetic field from coil 1 to coil 2.
- The combination of the two coils is called a transformer.

- The coil connected to the source is called the primary winding (or the primary)
- The coil connected to the load is called the secondary winding (or the secondary).

A decorative graphic consisting of a large, light blue arc that starts from the top left and curves towards the bottom right. A wedge-shaped area, also in light blue, is positioned on the right side of the slide, pointing towards the center.

Ideal transformer

- An ideal transformer has
 - no losses
 - no leakage flux and
 - its core is infinitely permeable.
- The mutual flux Φ_m is confined to the iron.
- $I_1 = 0$ on no load



Ideal Transformer

Properties of an ideal transformer

A decorative graphic consisting of a large, light blue arc that starts from the top left and curves towards the bottom right. A smaller, darker blue triangle is positioned at the end of this arc, pointing towards the bottom right corner of the slide.

E.m.f. equation and voltage ratio

- With the primary connected to an ac source V_1 , an alternating flux Φ_m is produced in the core.

Why?

- Let the flux be expressed as

$$\Phi_m = \Phi_{\max} \sin \omega t \text{ webers}$$

- The induced emf e_1 in the primary is given by

$$e_1 = \frac{d}{dt}(N_1 \Phi_m) = N_1 \frac{d}{dt}(\Phi_{\max} \sin \omega t)$$

or

$$e_1 = \omega N_1 \Phi_{\max} \cos \omega t$$

- Hence

$$e_1 = \omega N_1 \Phi_{\max} \sin(\omega t + 90^\circ)$$

$$E_1 = \frac{\omega N_1 \Phi_{\max}}{\sqrt{2}} = \frac{2\pi f N_1}{\sqrt{2}} \Phi_{\max}$$

or

$$E_1 = 4.44 f N_1 \Phi_{\max}$$

● Similarly

$$E_2 = \frac{\omega N_2 \Phi_{\max}}{\sqrt{2}} = \frac{2\pi f N_2 \Phi_{\max}}{\sqrt{2}}$$

$$E_2 = 4.44 f N_2 \Phi_{\max}$$

● Hence

$$\frac{E_1}{E_2} = \frac{N_1}{N_2}$$

- The ratio

$$a = N_1/N_2$$

is called the turns ratio.

- A step-up transformer has

$$a < 1$$

- A step-down transformer has

$$a > 1.$$

- In an ideal transformer, the applied voltage V_1 and the induced voltage E_1 must be identical. Why?

- Hence we may write

$$V_1 = E_1$$

- And

$$V_1 = 4.44 f N_1 \Phi_{\max}$$

- The above e.m.f. equation indicates that for a given
 - frequency,
 - number of turns and
 - voltage,
- the peak flux must remain constant. And so?

Current ratio and power equation

- On no load $I_1 = 0$.
- Now if a load is connected across the secondary terminals (i.e. switch S is closed) current I_2 flows through the load.
- This current produces mmf $N_2 I_2$ which if it acted alone would by Lenz's law, cause the mutual flux to reduce.

- Since when V_1 is fixed the flux Φ_{max} is also fixed, the primary develops mmf $N_1 I_1$ which is such that

$$N_1 I_1 = N_2 I_2$$



$$I_1 = I_2 / a$$

- In ideal transformer the secondary voltage

$$V_2 = E_2 \text{ Why?}$$

- V_2 remains constant since E_2 is constant when the Φ_{max} is fixed.

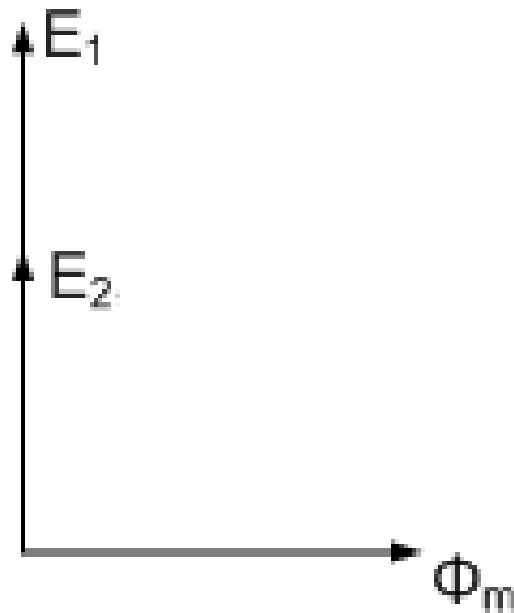
- It can be deduced from the above equations that

$$\frac{E_1}{E_2} = \frac{V_1}{V_2} = \frac{N_1}{N_2} = \frac{I_2}{I_1}$$

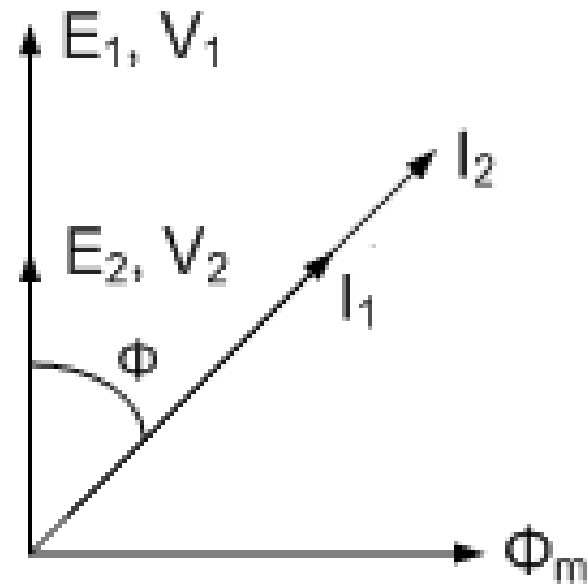
- Hence

$$V_1 I_1 = V_2 I_2$$

- That is there are no reactive and active losses in an ideal transformer.



(a) No load



(b) Load is resistive inductive

Phasor Diagram of an Ideal Transformer

Example 1

An ideal transformer having 90 turns on the primary and 2250 turns on the secondary is connected to a 200-V, 50-Hz source. The load across the secondary draws a current of 2 A at a power factor of 0.80 lagging.

Calculate

- (a) the rms value of the primary current
- (b) the flux linked by the secondary winding
- (c) Draw the phasor diagram.

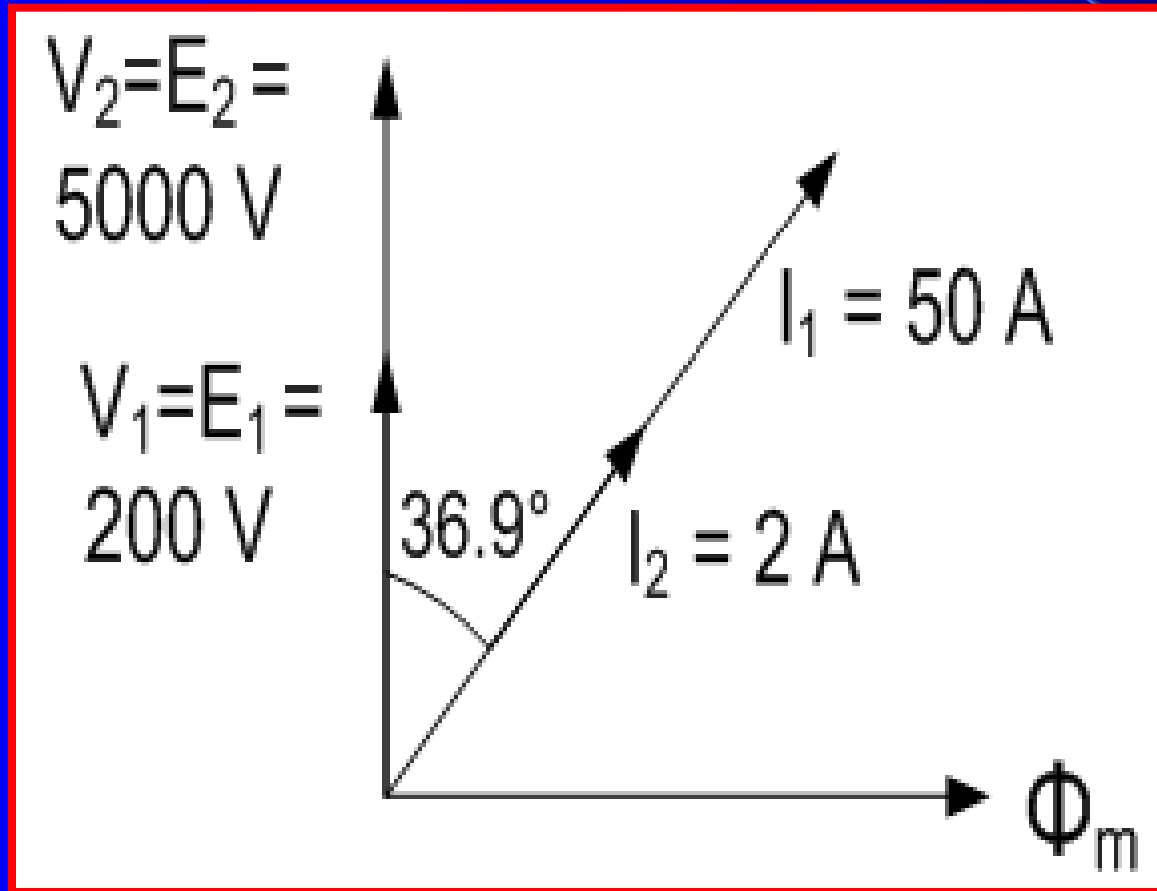
Solution

$$(a) \quad N_1 I_1 = N_2 I_2 \quad \blacktriangleright \quad 90 I_1 = 2250 \times 2 \quad \blacktriangleright \quad I_1 = 50 \text{ A}$$

$$(b) \quad \Phi_{\max} = \frac{V_1}{4.44 f N_1} = \frac{200}{4.44 \times 50 \times 90} = 0.01 \text{ Wb}$$

$$(c) \quad E_2 = \left(\frac{N_2}{N_1} \right) E_1 = 5000 \text{ V}$$

The phase angle between V_2 and I_2 is $\cos^{-1}(0.8) = 36.9^\circ$.



Example 2

A 200-kVA, 6600-V / 400-V, 50-Hz 1-ph transformer has 80 turns on the secondary. Calculate

- (a) the approximate values of the primary and secondary full-load currents
- (b) the approximate number of primary turns and
- (c) the maximum value of the flux.

Solution

(a) Full - load primary current

$$\cong \frac{200 \times 1000}{6600} = 30.3 \text{ A}$$

Full - load secondary current

$$= \frac{200 \times 1000}{400} = 500 \text{ A}$$

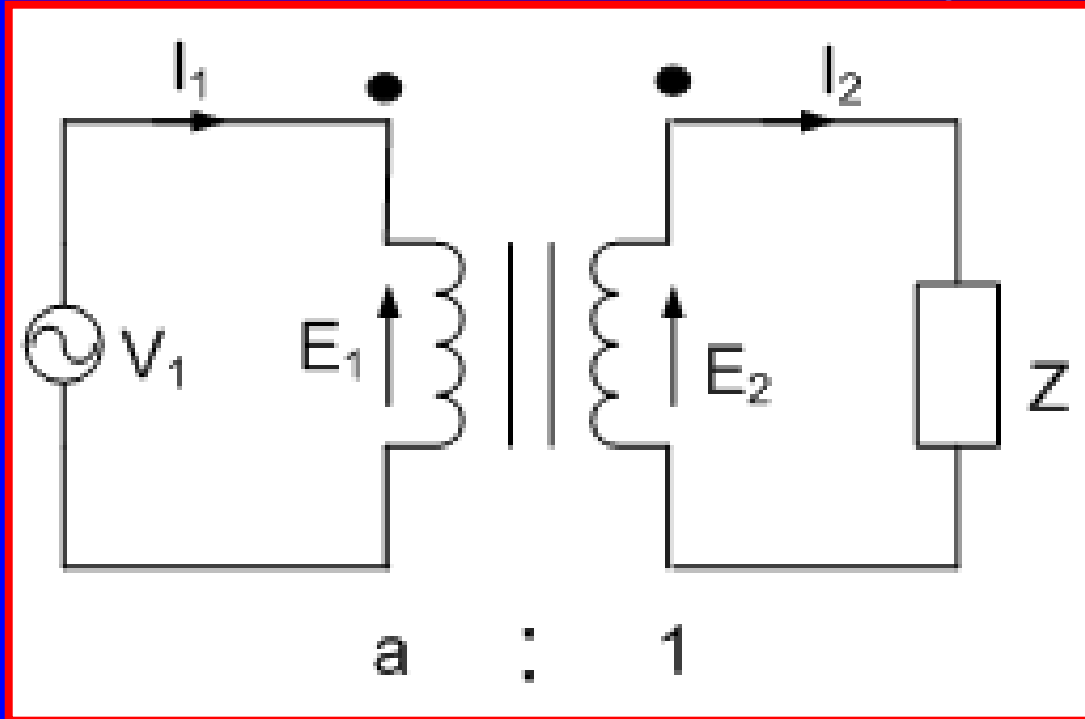
$$(b) \quad N_1 \cong \frac{80 \times 6600}{400} = 1320$$

$$(c) \quad \Phi_{\max} = \frac{V_2}{4.44 f N_2}$$

$$= \frac{400}{4.44 \times 80 \times 50} = 0.0225 \text{ Wb}$$

Impedance ratio

- Consider the circuit below.



- The impedance seen by the source is

$$Z_e = V_1 / I_1 = E_1 / I_1 = a E_2 / (I_2 / a) = a^2 E_2 / I_2 = a^2 Z$$

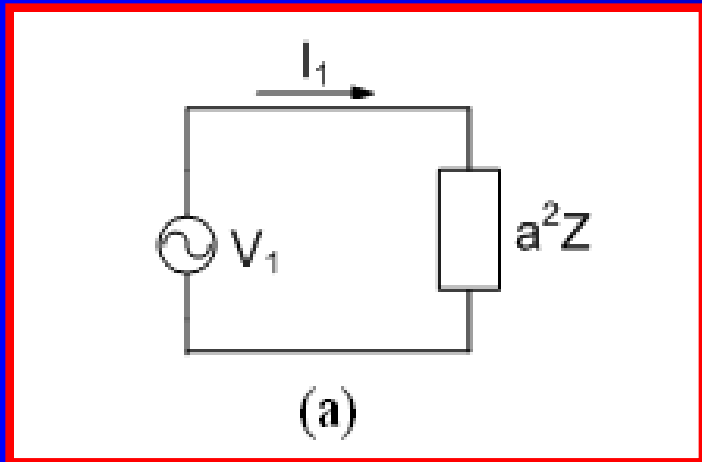
Equivalent circuits of an ideal transformer

$$\frac{V_1}{I_1} = a^2 Z$$



$$V_1 = [a^2 Z] I_1$$

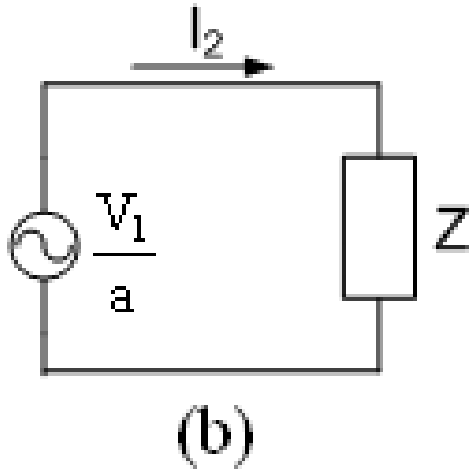
- According to the above equation, the transformer can be represented by the following equivalent circuit:



- We may also write

$$I_2 Z = E_2 = \frac{E_1}{a} = \frac{V_1}{a}$$

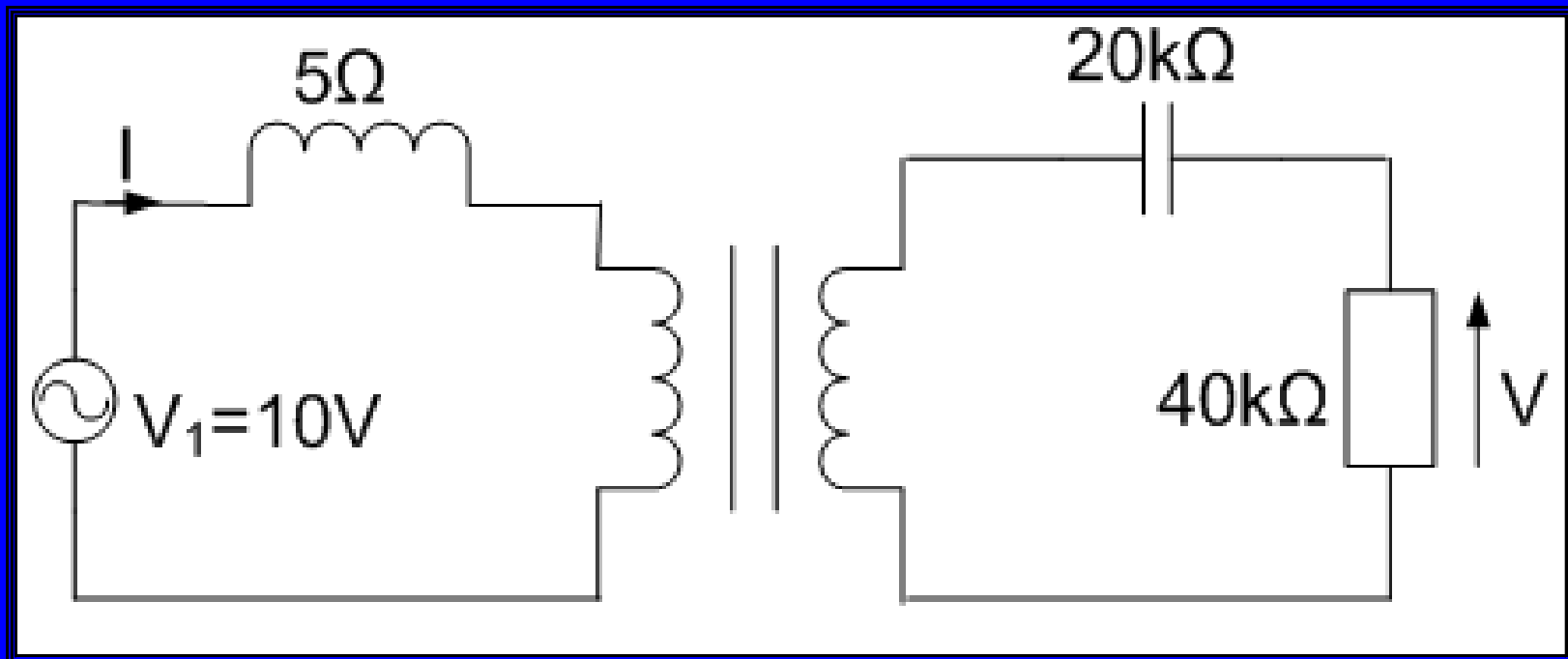
and represent it by the circuit below.



- The circuit (a) is known as the equivalent circuit referred to the primary side and
- The circuit (b) as the equivalent circuit referred to the secondary side.

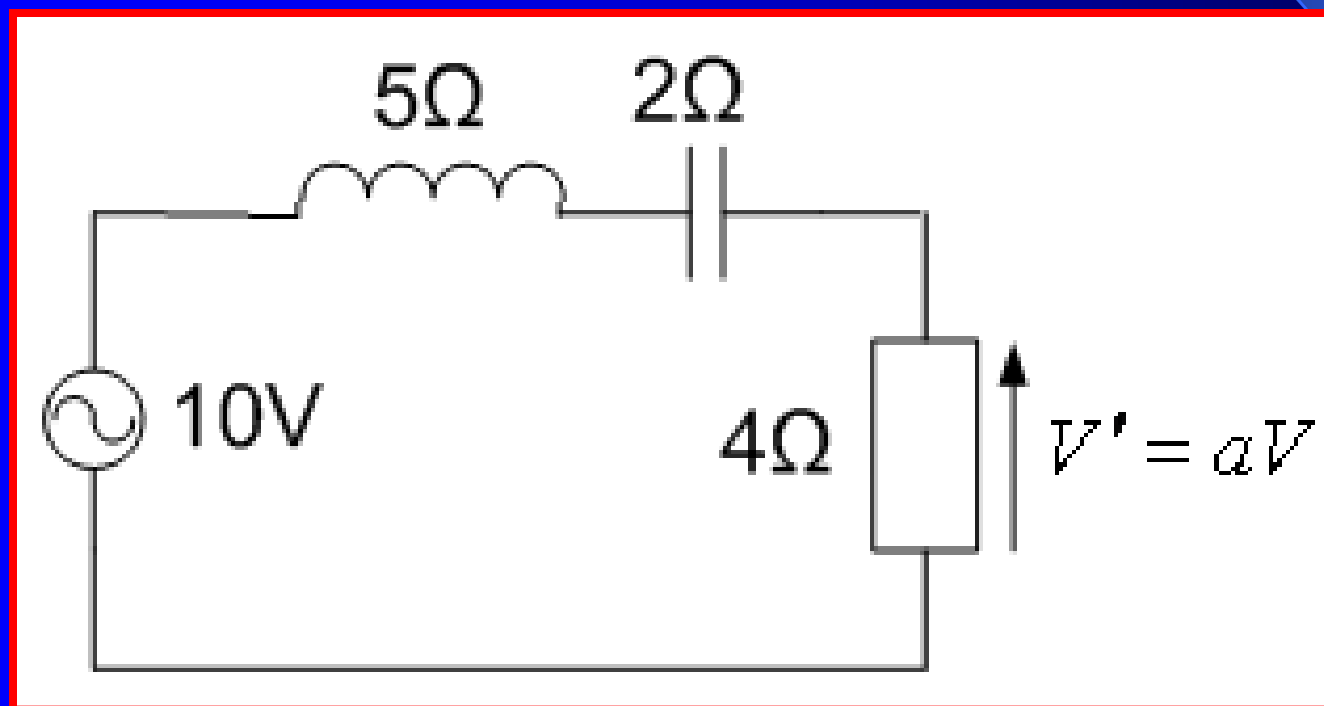
Example 3

Calculate the voltage V and current I in the figure below, knowing that the ideal transformer has a primary to secondary turns of 1:100 (i.e. $a = 1/100$).



Solution:

- We can avoid solving simultaneous equations by shifting all impedances to the primary side



$$Z_e = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{4^2 + 3^2} = 5\Omega$$

$$I = \frac{V_1}{Z_e} = \frac{10}{5} = 2\text{ A}$$

$$V' = IR = 2 \times 4 = 8\text{ V}$$

$$\text{The actual voltage } V = \frac{V'}{a} = 100 \times 8 = 800\text{ V}$$

Practical single-phase transformer

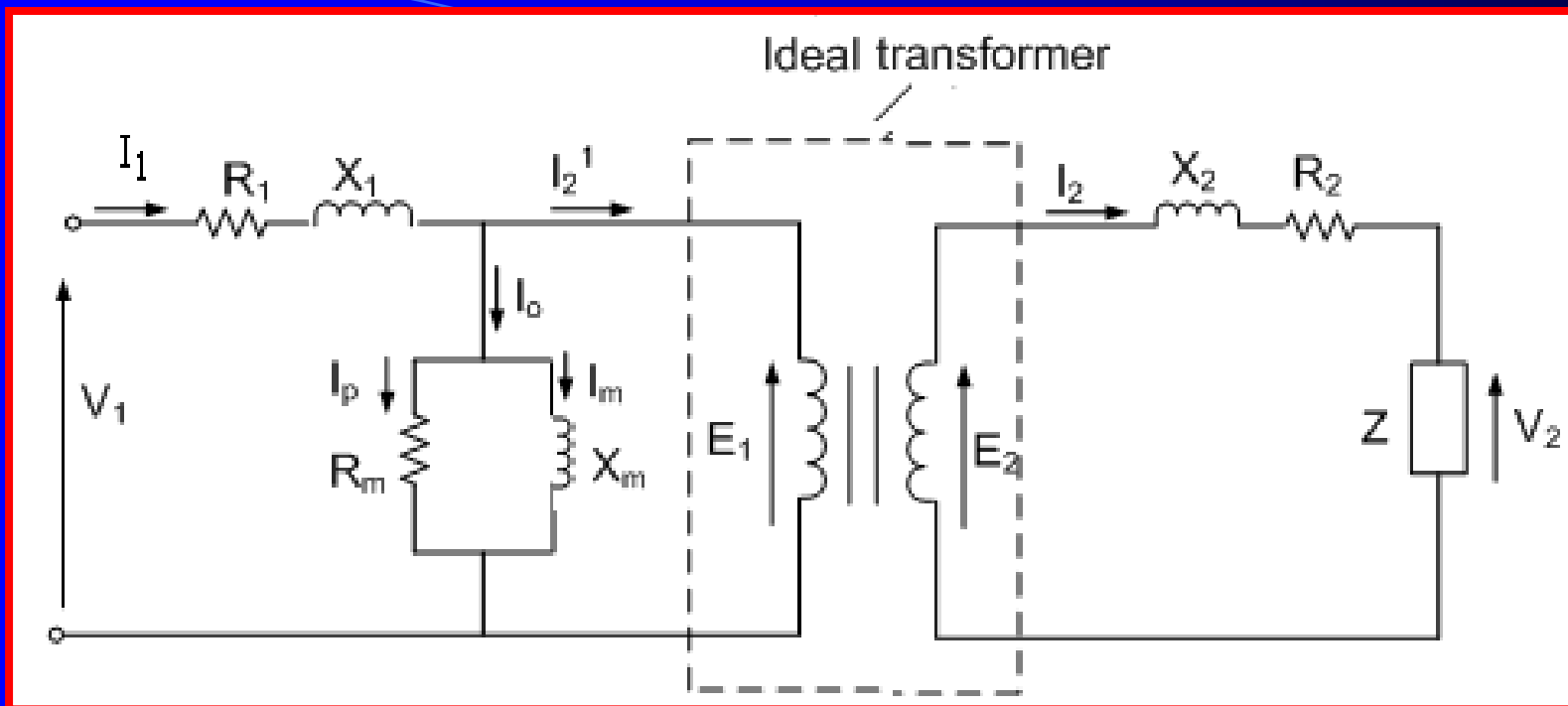
Introduction

- The windings of a practical transformer have
 - both resistance
 - and leakage inductance (This results from leakage flux).

- The core is also imperfect:
 - it has a core loss and
 - finite permeability (This means *mmf* is required to produce flux in the core).
- The core loss has two components:
 - hysteresis loss and
 - eddy current loss.

Equivalent circuit of a practical transformer

- The equivalent circuit is obtained by inserting impedances in the equivalent circuit of the ideal transformer to account for the imperfections of the actual transformer.

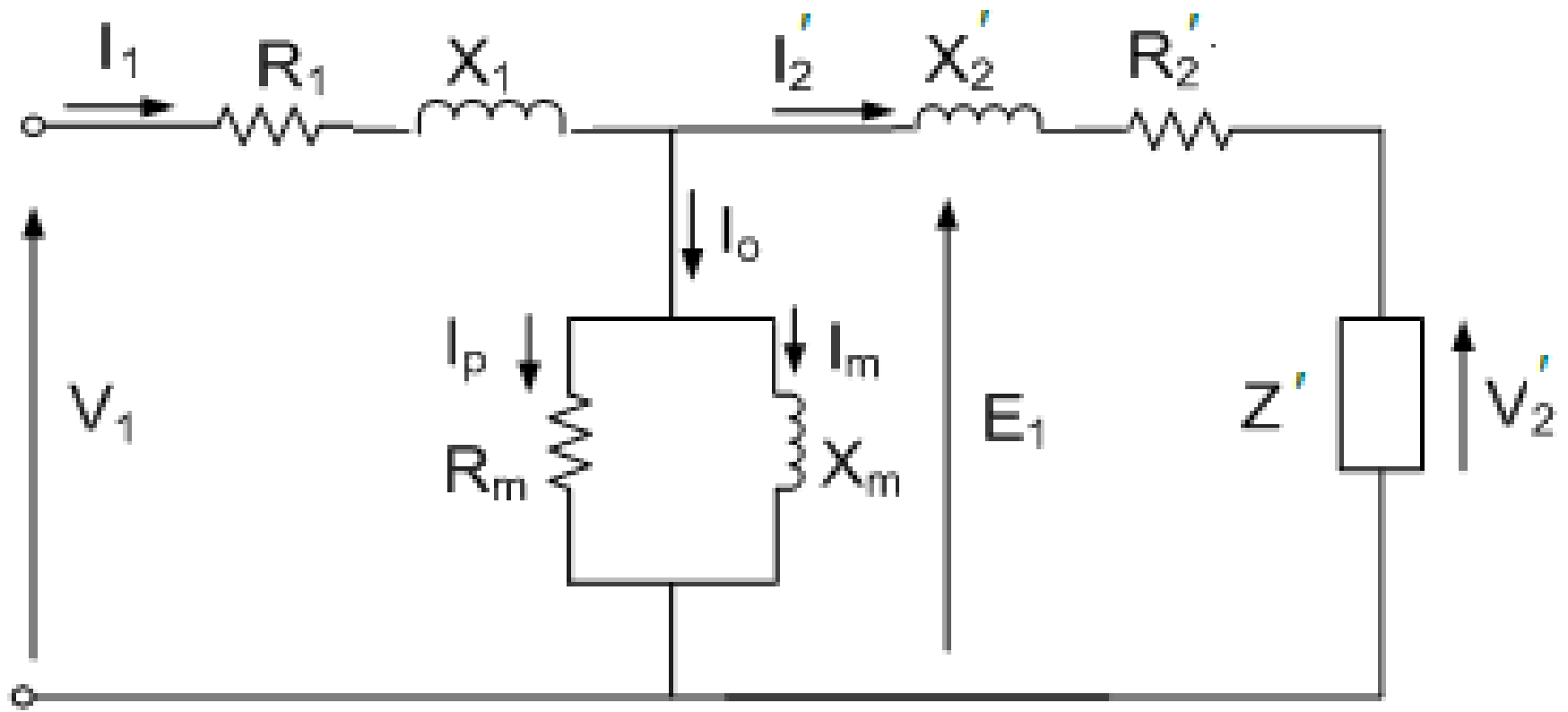


Equivalent circuit of a practical transformer

Ideal transformer equations still apply:

$$I_2' N_1 = N_2 I_2 \text{ and } E_1 / E_2 = N_1 / N_2$$

- R_1 and R_2 are resistances of the primary and secondary windings.
- X_1 and X_2 are the leakage reactances.
- The reactance X_m is such that it takes a reactive current I_m (i.e. the magnetizing current) to magnetize the core.
- The core loss is accounted for by the resistor R_m which takes the component I_p of the primary current.



Exact equivalent circuit referred to the primary side

Let $a = N_1/N_2$ Then

$$X'_2 = (N_1/N_2)^2 X_2 = a^2 X_2,$$

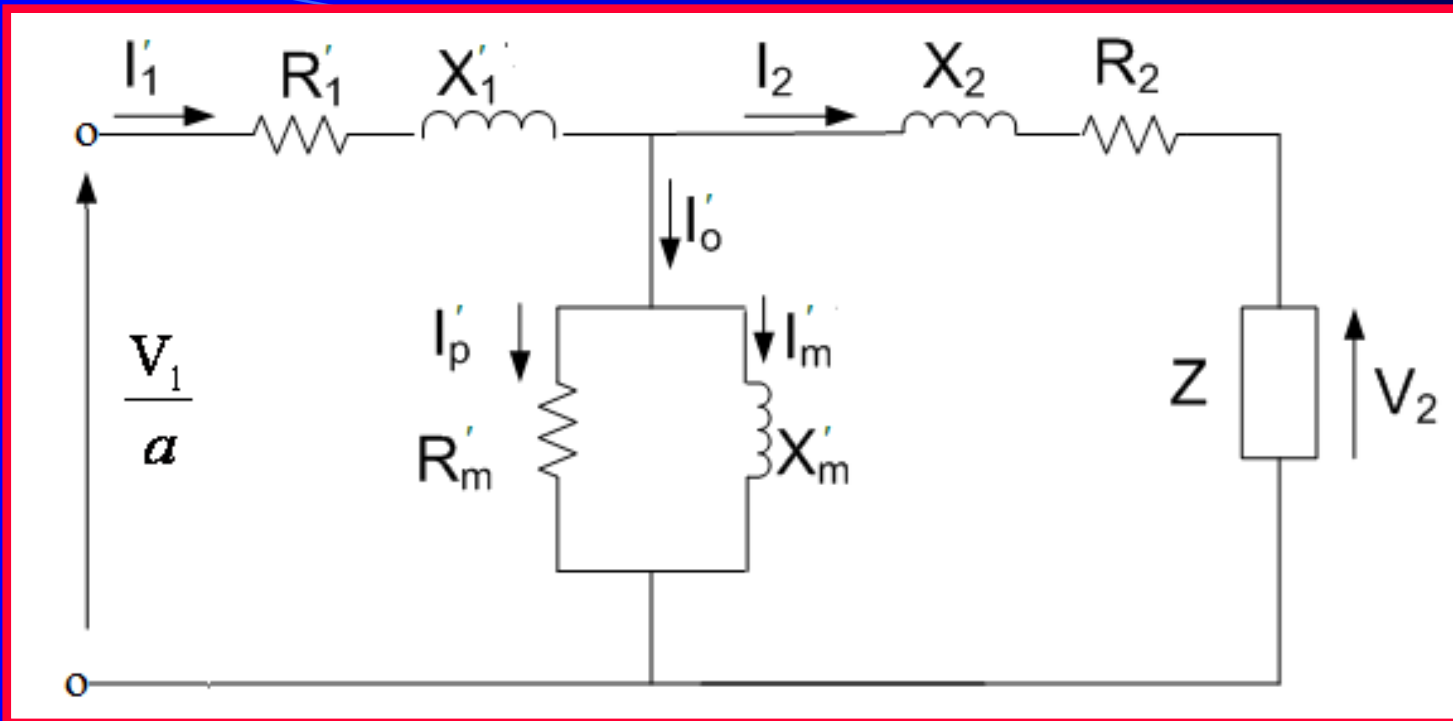
$$R'_2 = (N_1/N_2)^2 R_2 = a^2 R_2,$$

$$Z' = (N_1/N_2)^2 Z = a^2 Z$$

$$V'_2 = (N_1/N_2) V_2 = a V_2 \text{ and } I'_2 = I_2 / a$$

It is worth noting that for a practical transformer,

$$R'_2 \cong R_1 \quad \text{and} \quad X'_2 \cong X_1$$



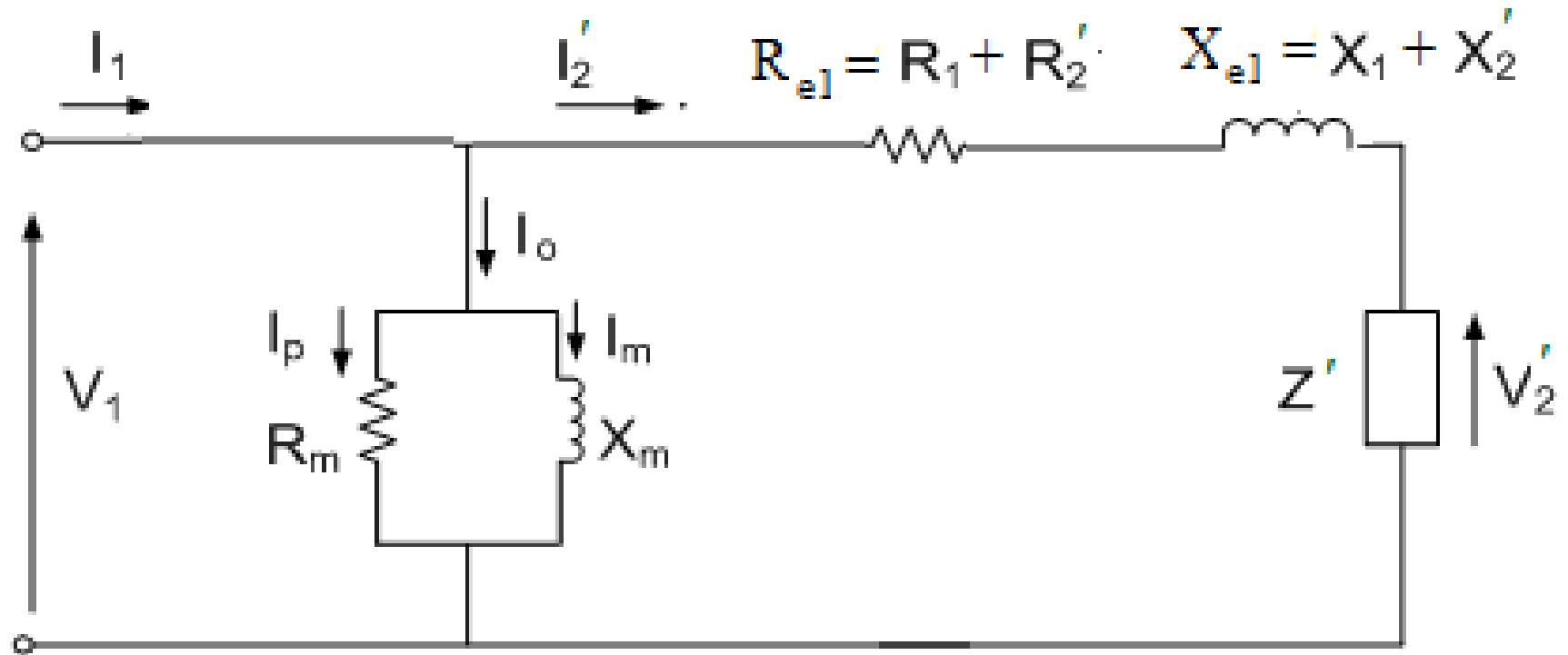
Exact equivalent circuit referred to the secondary side

$$R_1' = (N_2/N_1)^2 R_1 = R_1/a^2, \text{ etc}$$

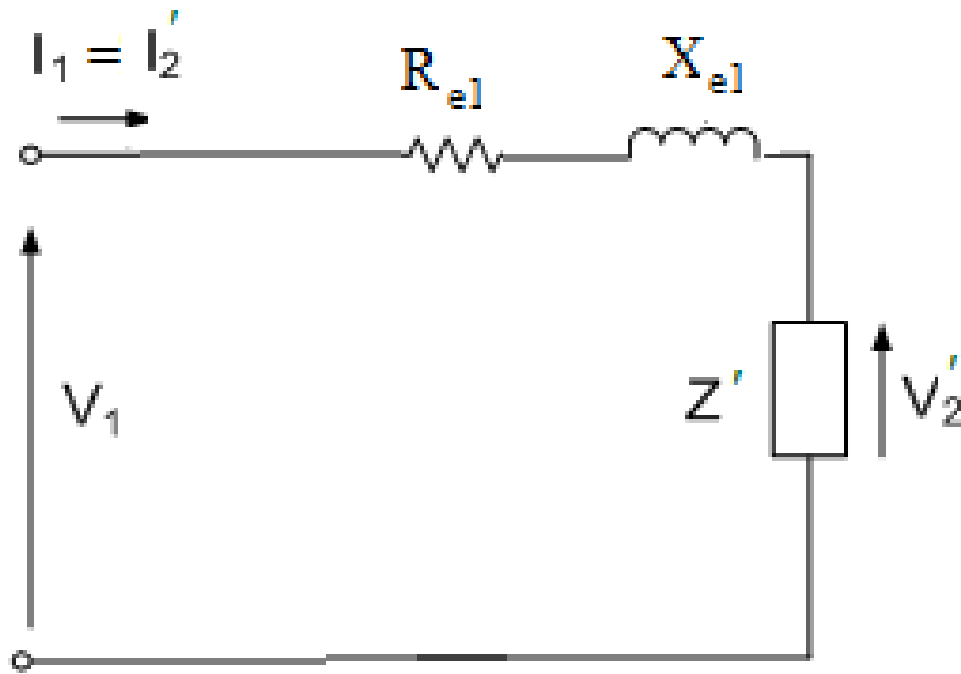
$$N_2 I_p' = N_1 I_p \quad \text{or} \quad I_p' = a I_p \text{ etc}$$

Approximate equivalent circuits

- The exact equivalent circuit of the transformer is too exact for most practical engineering applications.
- Consequently, we can simplify it to make calculations easier.
- The approximate equivalent circuits commonly used for power transformer calculations include the following:



The voltage drop in the primary leakage impedance due to the exciting or no load current I_o is neglected.



In the above circuit, the magnetizing branch is ignored entirely. This circuit is used in some applications such as voltage regulation.

Rating of transformers

- To keep the transformer temperature at an acceptable level, limits are set to both
 - the applied voltage (this determines the iron loss at a given frequency) and
 - the current drawn by the load (this determines copper loss).

- The limits determine the transformer's
 - rated voltage and
 - rated current
- The power rating of transformer
$$S_{\text{rated}} = \text{rated voltage} \times \text{rated current}$$

- It can be expressed in VA, kVA or MVA depending on the size of the transformer.
- The **rated kVA** (i.e. the rated power), **frequency** and **voltage** are always shown on the name plate.
- In large transformers, the corresponding currents are also shown.

- $\text{Rated kVA} = V_{1r} \times I_{1fl} \times 10^{-3}$
 $= V_{2r} \times I_{2fl} \times 10^{-3}$
- V_{1r} = rated primary voltage
- I_{1fl} = rated or full-load primary current
- V_{2r} = rated secondary voltage (It is the no load secondary voltage when the rated primary voltage is applied).
- I_{2fl} = rated or full-load secondary current

The turns ratio

- The turns ratio is given by $a = E_1/E_2$
- On no load, with V_{1r} applied, $E_1 \approx V_{1r}$ because $I_o Z_1$ is very small and $E_2 = V_{2r}$
- Thus $a \approx V_{1r} / V_{2r}$
- The turns ratio is calculated using the above equation, i.e. $a = V_{1r} / V_{2r}$
- We also note that $I_{1fl} = I_{2fl} / a$

Example 4

A 2200/200 V transformer takes 1 A at the high voltage side on no load at a power factor of 0.385 lagging. A load taken from the secondary of the transformer results in a current,

$I'_2 = 4.55 \angle -\cos^{-1}(0.8) \text{ A}$ with the input voltage as the reference phasor.

Consider the transformer to be represented by approximate circuit having the magnetizing branch placed across the primary terminals. Find

- (a) the actual secondary current.
- (b) the actual primary current

Solution

The secondary current

$$I_2 = \frac{N_1 I'_2}{N_2} = \frac{V_{1r} I'_2}{V_{2r}} = \frac{2200 \times 4.55}{200} = 50.05 \text{ A}$$

The primary current

$$\begin{aligned} \mathbf{I}_1 &= \mathbf{I}_0 + \mathbf{I}'_2 \\ &= 1 \angle -\cos^{-1}(0.385) + 4.55 \angle -\cos^{-1}(0.8) \\ &= (1 \times 0.385) + (4.55 \times 0.8) \\ &\quad + j[(1 \times 0.923) + (4.55 \times 0.6)] \\ &= 3.653 + j4.025 = 5.44 \angle -\cos^{-1}(0.74) \end{aligned}$$

Example 5

A single-phase transformer operates from a 230-V supply. It has an equivalent resistance of $0.1\ \Omega$ and an equivalent leakage reactance of $0.5\ \Omega$ referred to the primary. The secondary is connected to a coil having a resistance of $200\ \Omega$ and a reactance of $100\ \Omega$. Calculate the secondary terminal voltage. The secondary winding has 4 times as many turns as the primary.

Solution

Ignore the magnetizing branch.

The load impedance is $\mathbf{Z} = (200 + j100)\Omega$

Refer to the primary side:

$$\begin{aligned}\mathbf{Z}' &= \left(\frac{N_1}{N_2}\right)^2 \mathbf{Z} = \left(\frac{1}{4}\right)^2 (200 + j100) \\ &= 12.5 + j6.25 \Omega\end{aligned}$$

$$\begin{aligned}\mathbf{Z}_T &= \mathbf{Z}_{e1} + \mathbf{Z}' = (0.1 + j0.5) + (12.5 + j6.25) \\ &= 12.6 + j6.75\end{aligned}$$

Use voltage divider rule to find the magnitude of the voltage V'_2 :

$$|V'_2| = \frac{230 \times |12.5 + j6.25|}{|12.6 + j6.75|}$$

$$= \frac{230 \times 13.9754}{14.2941} = 224.8719 \text{ volts}$$

The actual load voltage is

$$V_2 = \frac{V'_2}{a} = 4 \times 224.8719 = 899 \text{ volts}$$

Definition of per-unit impedances

- Expressing impedances in per unit greatly simplifies the calculation of currents and voltages in power systems.
- The primary and secondary leakage impedances Z_1 and Z_2 of transformers are therefore usually expressed in per unit.

- The impedances in per unit are given by

$$\hat{Z}_1 = \frac{Z_1}{Z_{1base}} \quad \text{and} \quad \hat{Z}_2 = \frac{Z_2}{Z_{2base}}$$

- Where

$$Z_{1base} = \frac{V_{1r}}{I_{1fl}} = \frac{V_{1r}^2}{S_{rated}} \quad \text{and} \quad Z_{2base} = \frac{V_{2r}}{I_{2fl}} = \frac{V_{2r}^2}{S_{rated}}$$

- The impedances are said to be expressed in per unit with reference to
 - a base voltage V_{1r} for the primary and V_{2r} for the secondary, and
 - a base power S_{rated}
- The total impedance of the transformer in per unit $\hat{Z}_e = \hat{Z}_1 + \hat{Z}_2$

Example 6

A single-phase transformer that is rated 3000 kVA, 69 kV / 4.16 kV, 60 Hz has an impedance of 8 %.

Calculate the total impedance of the transformer referred to

(a) the primary side

(b) the secondary side

Solution

$$(a) \quad Z_{1base} = \frac{V_{1r}^2}{S_{rated}} = \frac{69^2 \times 10^6}{3000 \times 10^3} = 1587 \, \Omega$$

$$Z_{e1} = \hat{Z}_{e1} \times Z_{1base} = 0.08 \times 1587 = 127 \, \Omega$$

$$(b) \quad Z_{2base} = \frac{V_{2r}^2}{S_{rated}} = \frac{4.16^2 \times 10^6}{3000 \times 10^3} = 5.7685$$

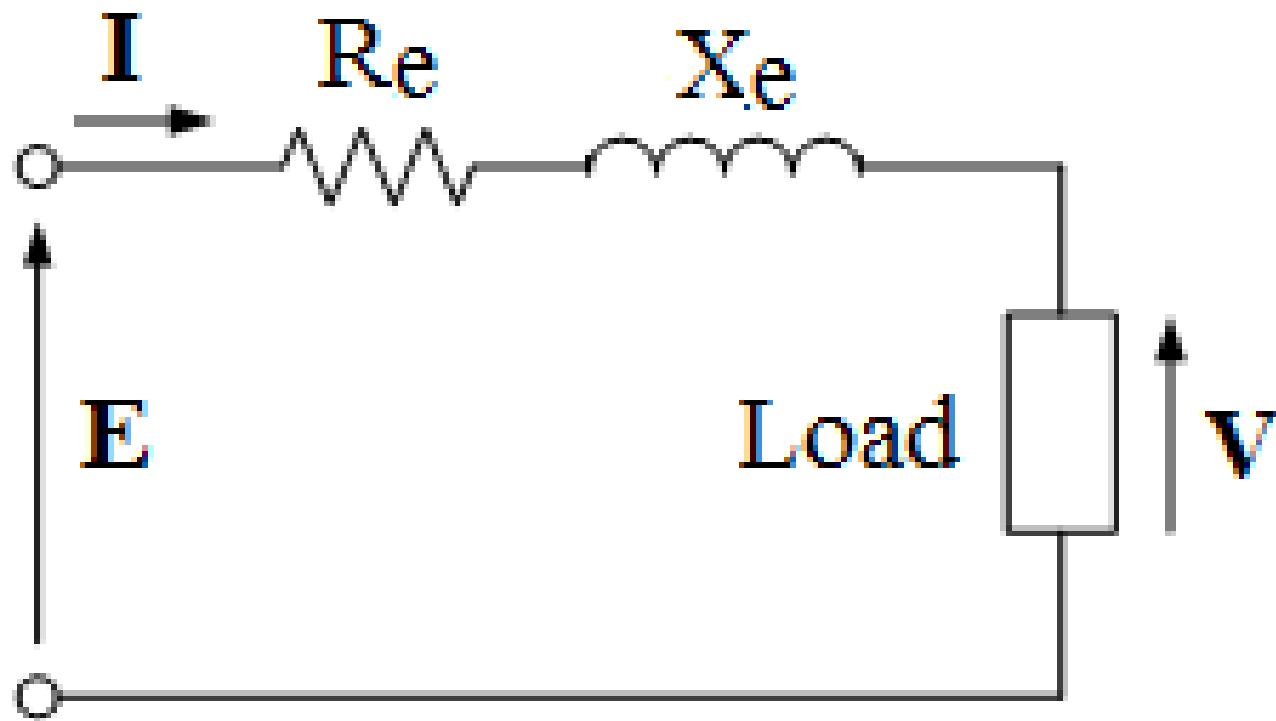
$$Z_{e2} = \hat{Z}_{e2} \times Z_{2base} = 0.08 \times 5.7685 = 0.46 \, \Omega$$

Voltage regulation

- The secondary terminal voltage at no load differs from the secondary voltage under load if the primary voltage is maintained constant.

- The voltage regulation or simply regulation of a transformer is the change in secondary voltage which occurs when the rated kVA output at a specified power factor is reduced to zero, with the primary voltage maintained constant.
- It is usually expressed as a percentage (called percentage regulation) or a fraction (in per unit) of the no-load terminal voltage.

- The approximate equivalent circuit with no magnetizing branch is used to calculate voltage regulation.
- The circuit may be either referred to the primary or the secondary side.
- The circuit in general form is shown below.



Where $V = V \angle 0^\circ$ and $I = I \angle -\phi$

- Load voltage = V
- When load is disconnected $V = E$
- Thus voltage regulation in volts is

$$\Delta V = E - V$$

- The voltage regulation in terms of the transformer parameters is given by

$$\Delta V = E - V = IR_e \cos \phi + IX_e \sin \phi$$

- The above equation applies to any load or load current I .

- The voltage regulation in per unit is given by

$$\varepsilon = \frac{E - V}{E} = \frac{I_{fl} R_e}{E} \cos \phi + \frac{I_{fl} X_e}{E} \sin \phi$$

- I_{fl} = the full-load current and $\cos \phi$ is the load power factor.
- The phase angle ϕ is positive when power factor is lagging and negative when power factor is leading.

- If the circuit is referred to the primary side, then

$$E = V_1, \quad V = V_2', \quad R_e = R_{e1} \quad \text{and} \quad X_e = X_{e1}$$

- Voltage regulation is then given by

$$\varepsilon = \frac{E - V}{E} = \frac{V_1 - V_2'}{V_1} \text{ pu}$$

- If the circuit is referred to the secondary side, then

$$E = V_{2n} = \frac{V_1}{a}, \quad V = V_2, \quad R_e = R_{e2} \quad \text{and} \quad X_e = X_{e2}$$

- Voltage regulation is then given by

$$\varepsilon = \frac{V_{2n} - V_2}{V_{2n}} \text{ pu}$$

- It can be shown that the two equivalent circuits give the same voltage regulation in per unit
- We also note that

$$V_2 = [1 - \varepsilon] V_{2n}$$



$$V_2 = [1 - \varepsilon] [V_1 / a]$$

- The voltage regulation is maximum for a given current or power when $\phi = \theta$ (impedance angle) where

$$\tan \theta = \frac{X_e}{R_e}; \sin \theta = \frac{X_e}{Z_e}$$

$$\cos \theta = \frac{R_e}{Z_e};$$

$$Z_e = \sqrt{R_e^2 + X_e^2}$$

$$\Delta V_{max} = I Z_e$$

Example 7

A 100-kVA 1-ph transformer has 400 turns on the primary and 80 turns on the secondary. The primary and secondary resistances are $0.3\ \Omega$ and $0.01\ \Omega$ respectively, and the corresponding leakage reactances are $1.1\ \Omega$ and $0.035\ \Omega$ respectively. The supply voltage is 2200 V.

Calculate:

- (a) the equivalent impedance referred to the primary circuit and
- (b) the voltage regulation and the secondary terminal voltage for full load having a power factor of
 - (i) 0.8 lagging and (ii) 0.8 leading
- (c) the maximum voltage regulation

Solution

$$(a) R_{e1} = R_1 + (N_1/N_2)^2 R_2$$
$$= 0.3 + 0.01 \times \left(\frac{400}{80} \right)^2 = 0.55 \, \Omega$$

$$X_{e1} = 1.1 + 0.035 \times \left(\frac{400}{80} \right)^2 = 1.975 \, \Omega$$

$$Z_{e1} = \left[0.55^2 + 1.975^2 \right]^{1/2} = 2.05 \, \Omega$$

$$(b) \quad (i) \cos \phi = 0.8 \quad \blacktriangleright \quad \sin \phi = 0.6$$

$$I_{1fl} = 100 \times 10^3 / 2200 = 45.45 \text{ A}$$

$$\begin{aligned} \varepsilon &= I_{1fl} \left(\frac{R_{e1} \cos \phi + X_{e1} \sin \phi}{V_1} \right) \\ &= 45.45 \left(\frac{0.55 \times 0.8 + 1.975 \times 0.6}{2200} \right) \\ &= 0.0336 \text{ pu} \end{aligned}$$

The secondary no load voltage

$$V_{2n} = V_1 \left(\frac{N_2}{N_1} \right) = 2200 \times \frac{80}{400} = 440 \text{ V}$$

The secondary terminal voltage

$$\begin{aligned} V_2 &= V_{2n} (1 - \varepsilon) = 440 (1 - 0.0336) \\ &= 425.2 \text{ V} \end{aligned}$$

$$\begin{aligned} b \quad (ii) \quad \varepsilon &= I_{1fl} \left(\frac{R_{e1} \cos \phi - X_{e1} \sin \phi}{V_1} \right) \\ &= 45.45 \left(\frac{0.55 \times 0.8 - 1.975 \times 0.6}{2200} \right) \\ &= -0.0154 \text{ pu} \end{aligned}$$

The secondary terminal voltage

$$V_2 = 440[1 - (-0.0154)] = 446.8 \text{ V}$$

(c) Maximum voltage regulation

$$\begin{aligned}\varepsilon_{\max} &= \frac{I_{1fl} Z_{e1}}{V_1} = \frac{45.45 \times 2.05}{2200} \\ &= 0.0424 \text{ pu.}\end{aligned}$$

Example 8

The primary and secondary windings of a 30-kVA, 6000-V / 230-V transformer have resistances of $10\ \Omega$ and $0.016\ \Omega$ respectively. The total reactance of the transformer referred to the primary is $23\ \Omega$. Calculate the percentage regulation of the transformer when supplying full load current at a power factor of 0.8 lag.

Solution

$$\begin{aligned} R_{e1} &= R_1 + (N_1/N_2)^2 R_2 \\ &= 10 + 0.016 \times (6000/230)^2 = 20.89 \, \Omega \end{aligned}$$

$$X_{e1} = 23 \, \Omega$$

$$I_{1fl} = (30 \times 10^3)/6000 = 5 \, \text{A}$$

$$\varepsilon = 5 \left(\frac{20.89 \times 0.8 + 23 \times 0.6}{6000} \right) = 0.0254 \, \text{pu}$$

Percent regulation = 2.54%

Efficiency

- The losses which occur in a transformer on load are composed of
 - Copper losses in primary and secondary windings, namely
$$I_1^2 R_1 + I_2^2 R_2 = I_1^2 R_{e1} = I_2^2 R_{e2}$$
 - Iron losses in the core due to hysteresis and eddy currents.

- The iron losses depend on the peak value of the mutual flux Φ_m and frequency.
- It is therefore independent of load current if voltage and frequency are constant.

● Let

■ P_c = the copper loss with full-load S_{rated} kVA in kW and

■ P_i = the iron losses (fixed loss) in kW

● Then at any load S given by

$S = x S_{rated}$ kVA at power factor $\cos\phi$, the total loss is $P_i + x^2 P_c$

- The efficiency of a transformer is calculated using this form of efficiency equation:

$$\eta = 1 - \frac{\text{losses}}{\text{losses} + \text{output}}$$

$$\eta = 1 - \frac{P_i + x^2 P_c}{x S_{rated} \cos \phi + P_i + x^2 P_c}$$

- For a given power factor, the efficiency is maximum when the load is such that

$$x = \sqrt{\frac{P_i}{P_c}}$$

- Or such that

the copper loss = the fixed loss or iron losses.

$$P_i = x^2 P_c$$

Example 9

The primary and secondary windings of a 500-kVA transformer have resistances of $0.42\ \Omega$ and $0.0011\ \Omega$ respectively. The primary and secondary voltages are 6600 V and 400 V respectively and the iron loss is 2.9 kW. Calculate the efficiency on full load at a power factor of 0.8.

$$I_{2fl} = \frac{500 \times 1000}{400} = 1250 \text{ A}$$

$$I_{1fl} = 500 \times \frac{1000}{6600} = 75.8 \text{ A}$$

Secondary copper loss on full load

$$= 1250^2 \times 0.0011 = 1720 \text{ W}$$

Primary copper loss on full load

$$= 75.8^2 \times 0.42 = 2415 \text{ W}$$

Total copper loss on full load,

$$P_c = 1720 + 2415 = 4135 \text{ W} = 4.135 \text{ kW}$$

Total loss on full load

$$= 4.135 + 2.9 = 7.035 \text{ kW}$$

Output power on full load at 0.8 pf

$$= 500 \times 0.8 = 400 \text{ kW}$$

$$\eta_{fl} = 1 - \frac{7.035}{400 + 7.035} = 0.9827 = 98.27\%$$

Example 10

Find the output, at which the efficiency of the transformer of example 9 is maximum and calculate its value assuming the power factor of the load to be 0.8.

Solution

$$x = \sqrt{P_i / P_c} = \sqrt{2.9 / 4.135} = 0.837$$

Therefore output at maximum efficiency,

$$S = 0.837 \times 500 = 418.5 \text{ kVA}$$

At maximum efficiency,

$$\text{total loss} = 2 \times 2.9 = 5.8 \text{ kW}$$

Output power at 0.8 pf is

$$P = 418.5 \times 0.8 = 334.8 \text{ kW}$$

$$\therefore \eta_{\max} = 1 - \frac{5.8}{5.8 + 334.8} = 0.983 = 98.30 \%$$

Example 11

A 400-kVA transformer has an iron loss of 2 kW and the maximum efficiency at 0.8 pf occurs when the load is 240 kW. Calculate the full-load copper loss.

Solution

Output kVA at maximum efficiency is

$$S = \frac{240}{0.8} = 300 \text{ kVA}$$

The fraction x of full-load kVA at which the efficiency is maximum

$$x = \frac{300}{400} = 0.75$$

$$\therefore P_c = \frac{P_i}{x^2} = \frac{2}{0.75^2} = 3.56 \text{ kW}$$

Open-circuit and short-circuit tests on a transformer

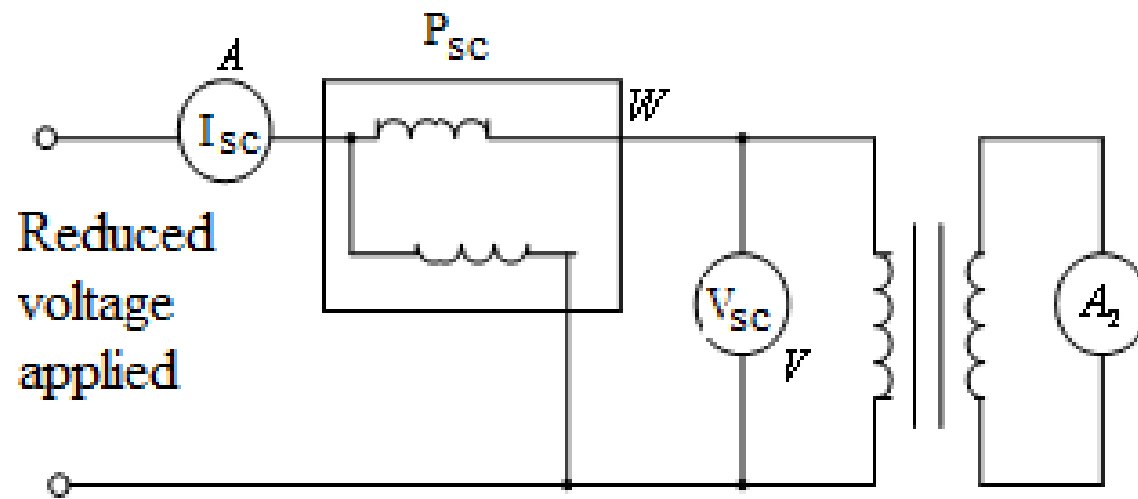
- These two tests enable us to calculate the following without actually loading the transformer:
 - the efficiency and
 - voltage regulation

Short-circuit test

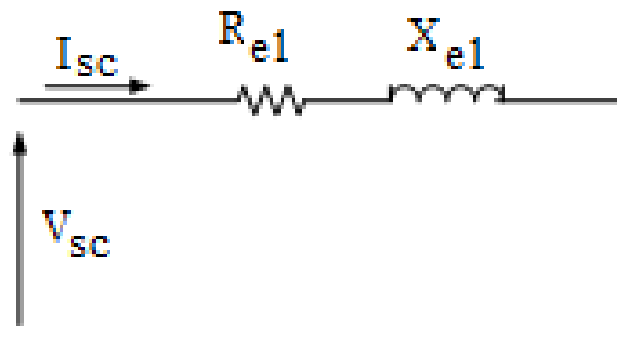
- This test is used to determine the leakage impedance.
- During this test, one winding is short-circuited and a reduced voltage V_{sc} applied to the other to cause rated current to flow.

- It is often convenient on the higher voltage transformers to supply the high-voltage winding, thus using a smaller current.
- V_{sc} which will be about 3-15 % of the rated value may also be more suitable for test facilities.

- The test and the equivalent circuits are shown below
- The magnetizing branch is neglected because its current under this condition is less than 1 % of the total.
- The circuit parameters obtained are referred to the side to which the test voltage is applied.



(a) Test circuit



(b) Equivalent circuit

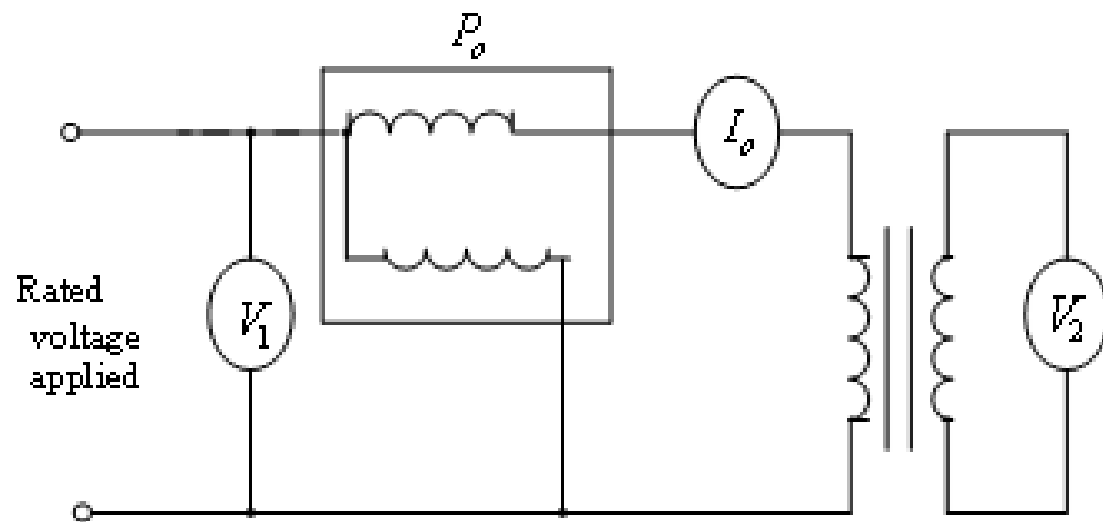
- The following quantities are measured by the instruments to make the required calculations:
the voltage V_{sc} , the current I_{sc} and the power P_{sc}
- The calculations are

$$(a) Z_{e1} = \frac{V_{sc}}{I_{sc}} \quad (b) R_{e1} = \frac{P_{sc}}{I_{sc}^2} \quad (c) X_{e1} = \sqrt{Z_{e1}^2 - R_{e1}^2}$$

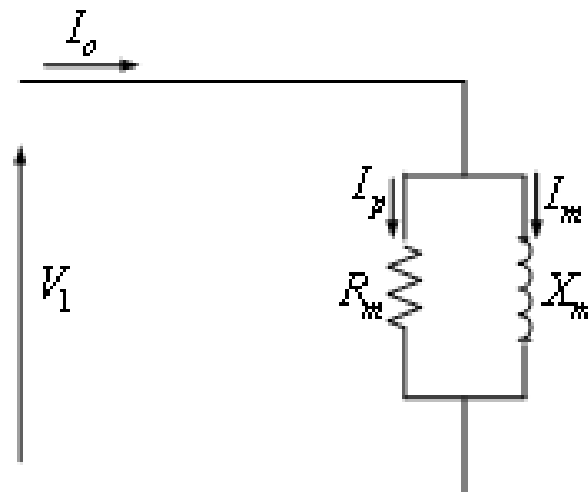
Open-circuit test or no load test

- During this test, one winding is open-circuited and rated voltage at rated frequency is applied to the other.
- Quite often the low-voltage side is supplied to reduce the test voltage required for safety reasons.

- Here also, the equivalent circuit parameters are referred to the side to which the test voltage is applied.
- The test and equivalent circuits are shown below



(a) Test circuit



(b) Equivalent circuit

- The following calculations can be made

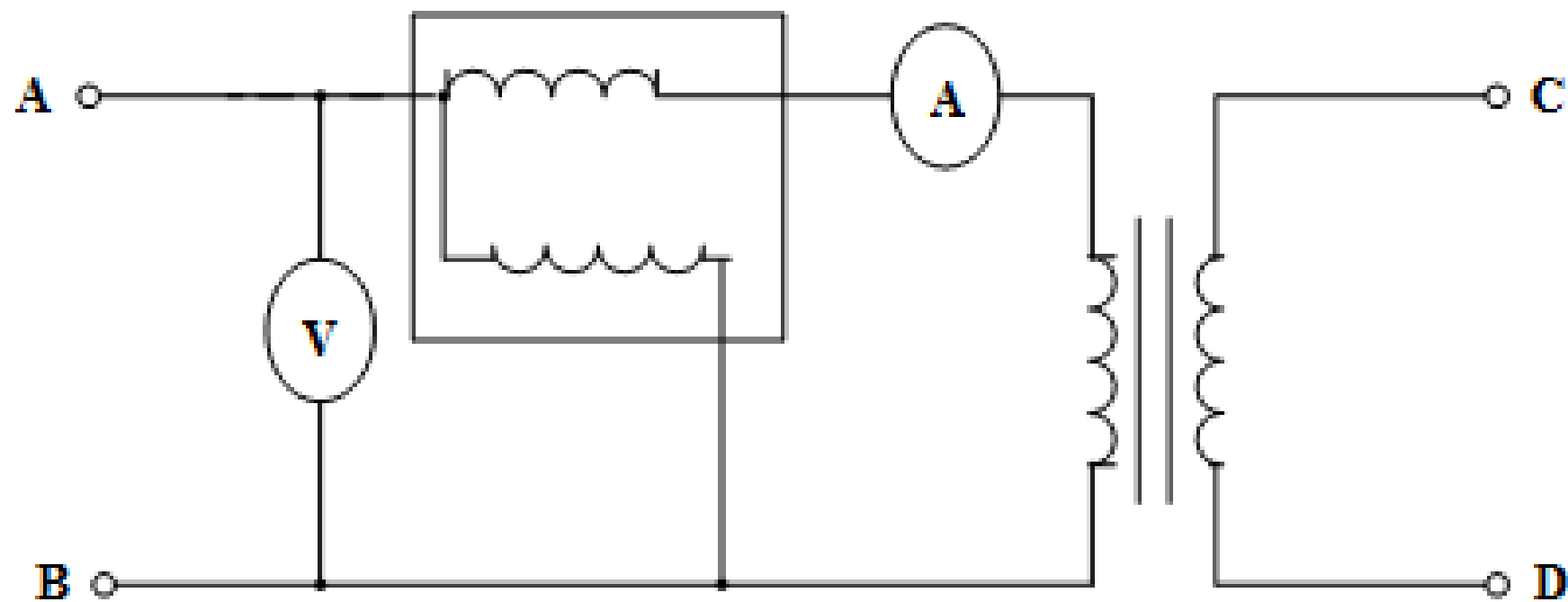
$$(a) I_p = \frac{P_o}{V_1} \quad (b) I_m = \sqrt{I_o^2 - I_p^2} \quad (c) R_m = \frac{V_1}{I_p}$$

$$(d) X_m = \frac{V_1}{I_m} \quad (e) a = \frac{V_1}{V_2}$$

- P_{sc} and P_o represent the full load copper loss and the core loss (or iron losses) respectively.
- They can be used directly to calculate efficiency.

Example 12

The circuit shown below was used in a test on a 3-kVA transformer. A variable voltage supply of fixed frequency was connected to terminals A and B and two tests were performed:



- (a) The voltage was raised to normal rated voltage and the meter reading were then 200 V, 24 W, 1.2 A
- (b) The terminals C and D were short-circuited and the voltage was raised until the transformer full-load current was flowing. The meter readings were then 6.4 V, 28 W, 15 A.

From the results of the tests, obtain:

- (i) the no-load current and its power factor
- (ii) the iron losses of the transformer at normal frequency and voltage
- (iii) the full-load copper loss
- (iv) the transformer resistances, R_{e1} and R_m and reactances X_{e1} and X_m
- (v) the efficiency of the transformer at full load at a power factor of 0.8

Solution

(i) No-load current $I_o = 1.2 \text{ A}$

No-load power factor

$$pf_o = \frac{P_o}{S_o} = \frac{24}{200 \times 1.2} = 0.1 \text{ lag}$$

(ii) iron losses $P_i = P_o = 24 \text{ W}$

(iii) full-load copper loss $P_c = P_{sc} = 28 \text{ W}$

(iv) Open-circuit test calculations:

$$I_p = \frac{P_o}{V_1} = \frac{24}{200} = 0.12 \text{ A}$$

$$I_m = \sqrt{I_o^2 - I_p^2} = \sqrt{1.2^2 - 0.12^2} = 1.19 \text{ A}$$

$$R_m = \frac{V_1}{I_p} = \frac{200}{0.12} = 1.67 \text{ k}\Omega$$

$$X_m = \frac{V_1}{I_m} = \frac{200}{1.19} = 168 \Omega$$

Short-circuit test calculations:

$$Z_{e1} = \frac{V_{sc}}{I_{sc}} = \frac{6.4}{15} = 0.43 \Omega$$

$$R_{e1} = \frac{P_{sc}}{I_{sc}^2} = \frac{28}{15^2}$$

$$= 0.12 \Omega$$

$$X_{e1} = \sqrt{Z_{e1}^2 - R_{e1}^2} = \sqrt{0.43^2 - 0.12^2} = 0.41 \Omega$$

$$(v) \quad \text{Efficiency} = 1 - \frac{24 + 28}{(3000 \times 0.8) + 24 + 28}$$

$$= 0.9788 \text{ pu} = 97.88 \%$$

Parallel operation of single-phase transformers

- The parallel connection of several transformers is widely used in electrical systems for the following reasons:
 - In many cases, there is no single transformer to transform that amount of power

- Frequently, as a result of load growth the installed transformer will be required to supply an output greater than its kVA capacity.

Additional transformers are then installed to run in parallel with the existing transformer.

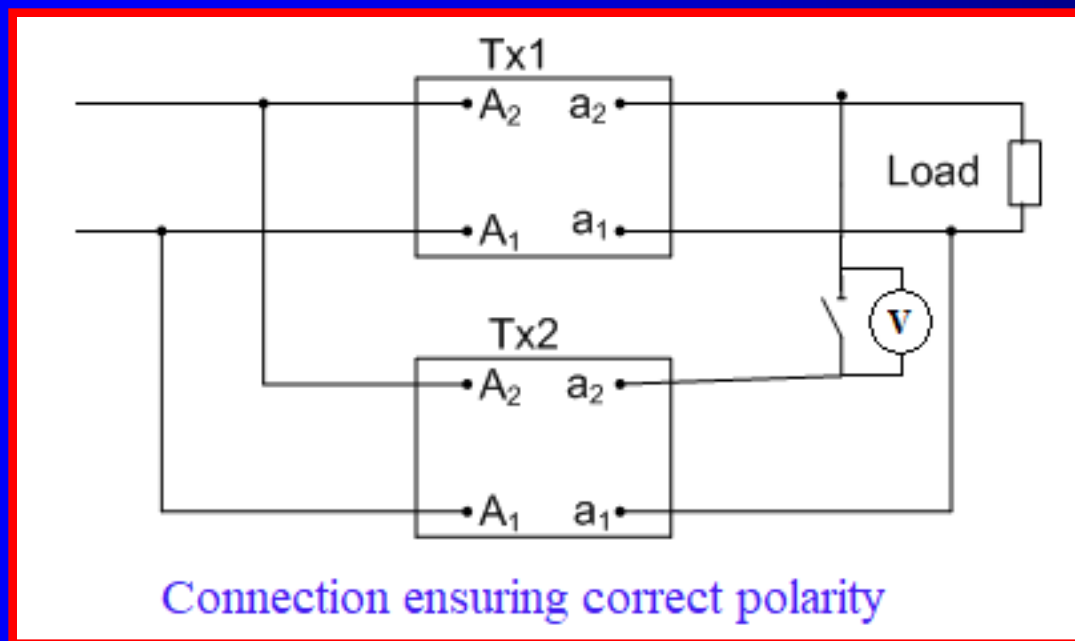
- It is sometimes found desirable to supply a load through two or more units in order to reduce the cost of the spare unit.

The spare unit is required to ensure continuity of service in case of damage.

- The following three conditions must be fulfilled when operating two or more single-phase transformers in parallel:

- The polarity must be the same
 - This means the induced secondary emfs in the transformers must have the same direction (or polarity).
 - The polarity can be either right or wrong.
 - A wrong polarity results in a severe short circuit.

- Thus this condition is essential.
- Terminals of the same markings are connected together to ensure correct polarity.



- If the polarity markings are either incorrect or not present, the polarity of the incoming transformer can be checked by connecting a voltmeter across the paralleling switch as shown above.
- Voltmeter reads almost zero if polarity is correct.

- The voltage ratio should be the same
 - This is to avoid no-load circulating current
 - And also over-loading on one transformer when the paralleled transformers are loaded.
 - This is desirable condition

- The per-unit impedances should have the same magnitude and the same angle
 - This is desirable condition.
 - When they are equal in magnitude, the transformers share kVA loads in proportion to their respective ratings.

- If both their magnitudes and angles are the same

- ❖ they will share kVA loads in proportion to their respective ratings and

- ❖ also the combined load kVA will be the algebraic sum of the kVA carried by each transformer.

- If the angles are different, the resultant kVA capacity of the paralleled group will be slightly smaller than the sum of their individual ratings if none should be overloaded.
- It is not very necessary that the angles should be the same.

Load sharing of parallel-connected transformers

- Consider two transformers operating in parallel sharing a total load S .
- If their voltage ratios are equal, then the load carried by each transformer will be calculated as follows:

$$S_1 = \frac{Z_2}{Z_1 + Z_2} S$$

$$; \quad S_2 = \frac{Z_1}{Z_1 + Z_2} S$$

Where

S = combined load

S_1 = transformer loading

S_2 = transformer 2 loading

Z_1 = transformer 1 leakage impedance

Z_2 = transformer 2 leakage impedance

The two impedances are all either referred to the primary or to the secondary side.

- The above equations hold for per-unit impedances provided that all are expressed with reference to a common base power.
- Use this to obtain a new per unit value with reference to a new base power:

$$Z_{pu}^{new} = \left(\frac{S_{base}^{new}}{S_{base}^{old}} \right) \times Z_{pu}^{old}$$

Example 14

A 500-kVA transformer having reactance of 8 % is to share a load of 450 kVA at power factor of 0.8 lagging with a 250-kVA transformer having a reactance of 8 %. Find the active power that will be supplied by the 250-kVA transformer.

Solution

If all the impedances are referred to a base power of 500 kVA, then only the impedance of the 250-kVA transformer will change.

$$Z_{pu}^{new} = \left(\frac{S_{base}^{new}}{S_{base}^{old}} \right) \times Z_{pu}^{old} = \left(\frac{500}{250} \right) \times 8 = 16 \%$$

$$\begin{aligned}
 S_{250} &= \frac{Z_{500}}{Z_{250} + Z_{500}} S \\
 &= \frac{8\angle 90^\circ}{16\angle 90^\circ + 8\angle 90^\circ} 450\angle -\cos^{-1}(0.8) \\
 &= \frac{8}{24} \times 450\angle -\cos^{-1}(0.8) \\
 &= 150 \text{ kVA} \angle -\cos^{-1}(0.8)
 \end{aligned}$$

Hence $S_{250} = 150 \text{ kVA}$ and $P_{250} = 150 \times 0.8 = 120 \text{ kW}$

Alternatively, we can use simple proportion since the two impedances are equal in magnitude and have the same angle.

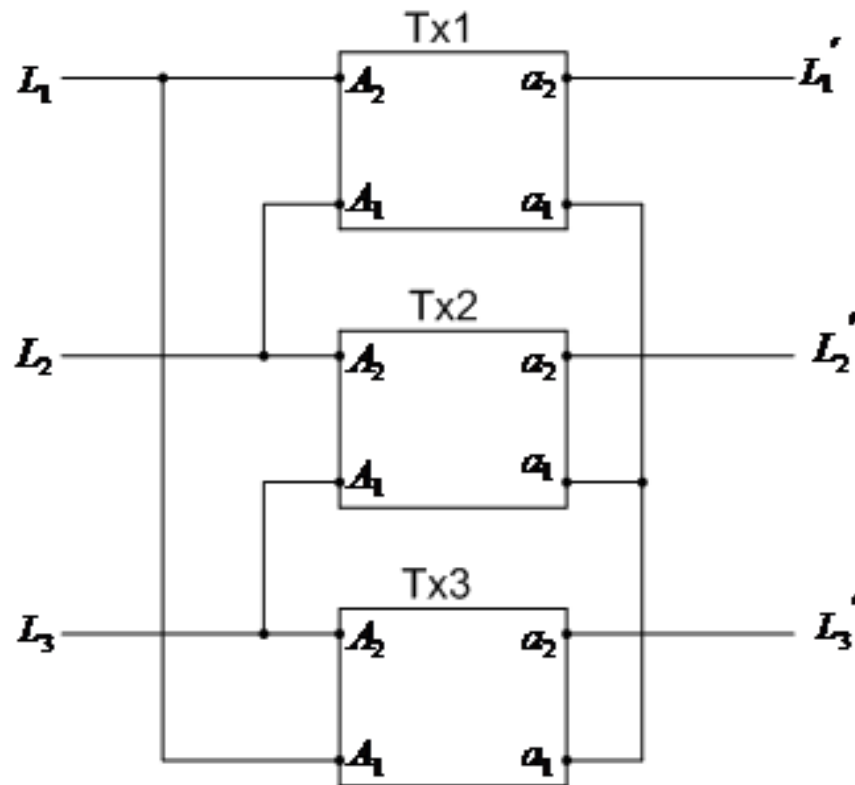
Three-phase transformers

- These are required to transform 3-phase power.
- The three-phase transformer may be either of the following:
 - A three-phase transformer bank
 - A three-phase transformer unit

A three-phase transformer bank

- This consists of three identical single-phase transformers having their windings externally connected for three-phase working.

- The single-phase transformers retain all their basic single-phase properties such as **current ratio**, **voltage ratio** and **the flux in the core**.
- The kVA capacity of the bank is the sum of their individual ratings.



Connection diagram of delta-star
three-phase transformer bank

A three-phase transformer unit

- This is a single unit of special construction for three-phase working.
- Modern large transformers are usually of the three-phase three-legged core type.

LV terminals

HV terminals

Cross-flow fans

Permitting a 50% increase in the rated power

Three-leg core

HV winding

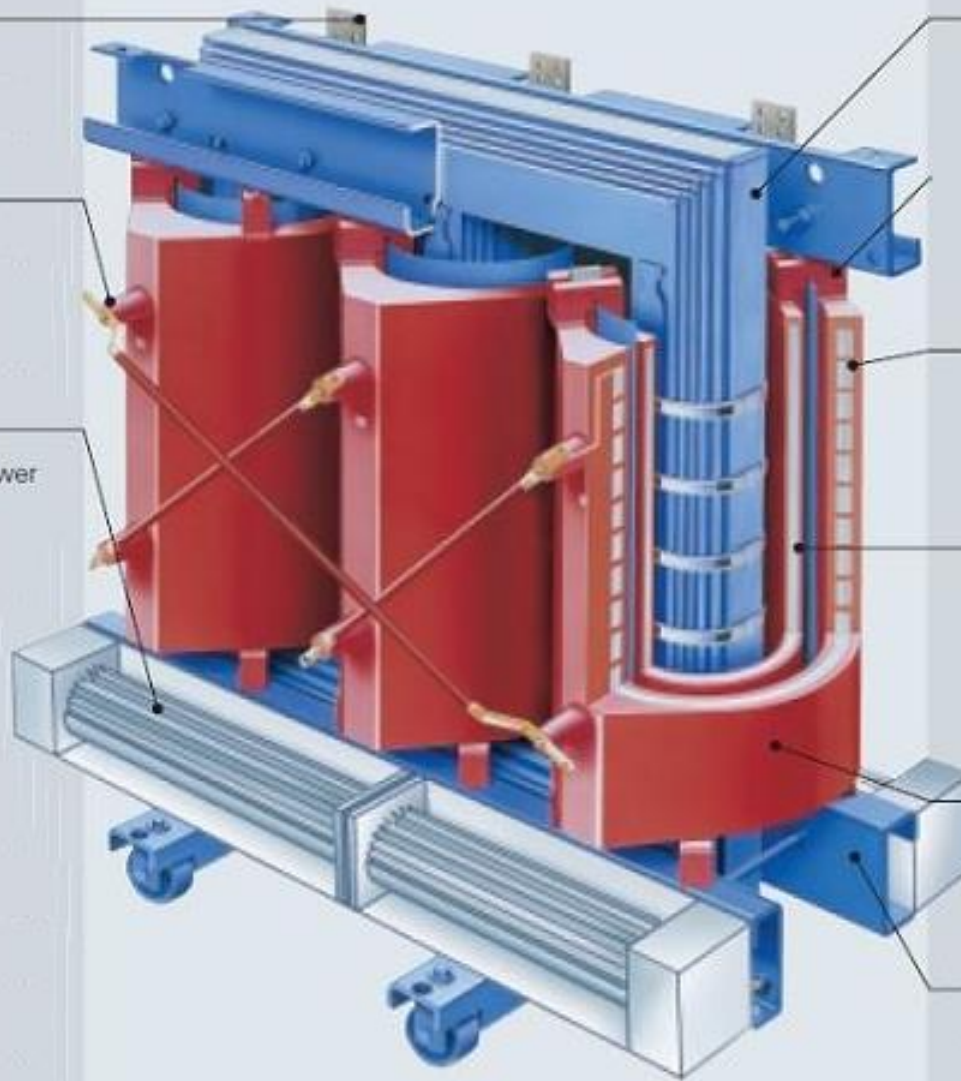
LV winding

Insulation:
Mixture of epoxy resin
and quartz powder

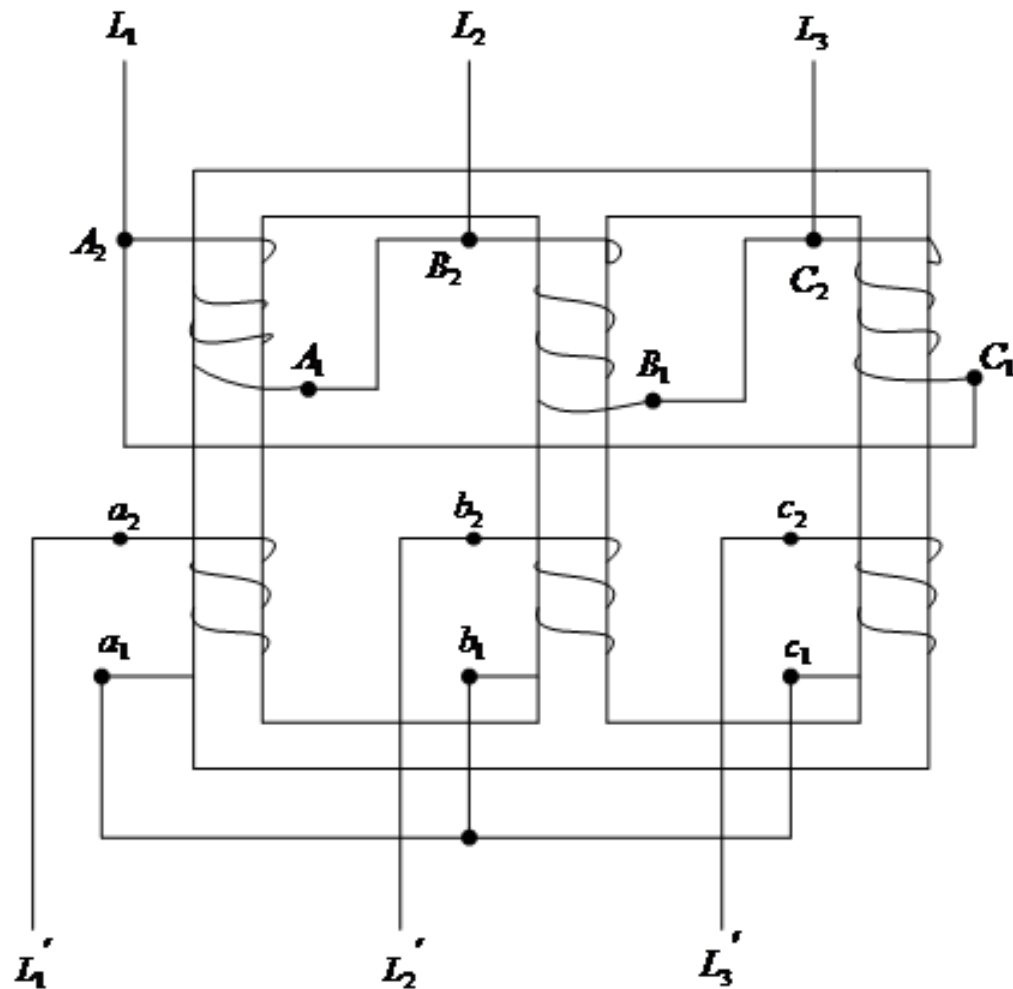
Makes the transformer
maintenance-free, moisture-proof,
tropicalized, flame-resistant and self-extinguishing

Clamping frame and truck

Rollers can be swung
around for lengthways or
sideways travel



Three-phase three-legged core type transformer



Connection diagram of delta-star
three-phase transformer unit

- A leg carries the primary and secondary windings of a phase.
- The windings are internally connected.
- For a given total capacity, 3-phase units are much cheaper in capital cost, lighter, smaller and more efficient.

Three-phase transformer winding connection

- The three windings, primary or secondary, can be connected in three different ways:
 - Star connection
 - Delta connection
 - Zigzag connection

- The three different winding connections give rise to several possible connection combinations: star-star, star-delta, star-zigzag, delta-star, delta-delta, etc
- The most common winding connections are the star and delta.

Star connection

- For this connection
 - phase current = line current and
 - phase voltage = line voltage $/\sqrt{3}$.
- The phase voltage is about 58 % of the line voltage.

- This enables the insulation of the winding to be reduced to a minimum for a given supply voltage
- It is the most economical connection for a high-voltage winding.

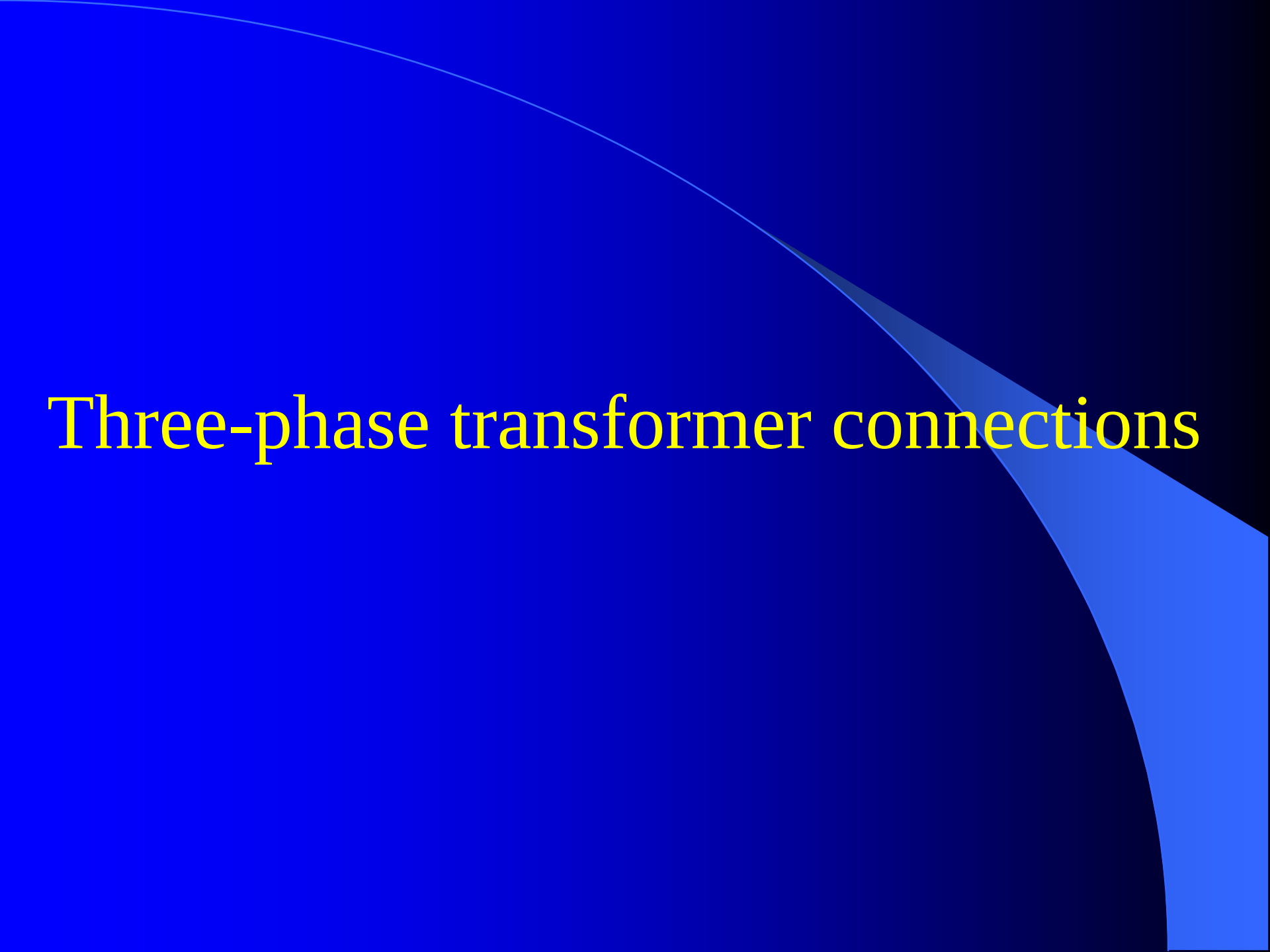
Delta connection

- For this connection
 - phase voltage = line voltage and
 - phase current = line current/ $\sqrt{3}$
- Because the phase = the line voltage the winding must be insulated for the full line voltage.

- More turns are also required.
- With very high voltages a saving of 10 % may be achieved by using star-connection rather than delta connection on account of insulation.
- The saving is small, however, at voltages below 11 kV.

- Because the phase current = line current/ $\sqrt{3}$, the winding cross-sectional area is 58 % of that required for the star connection.
- Therefore it is the most economical for low-voltage winding.

Three-phase transformer connections

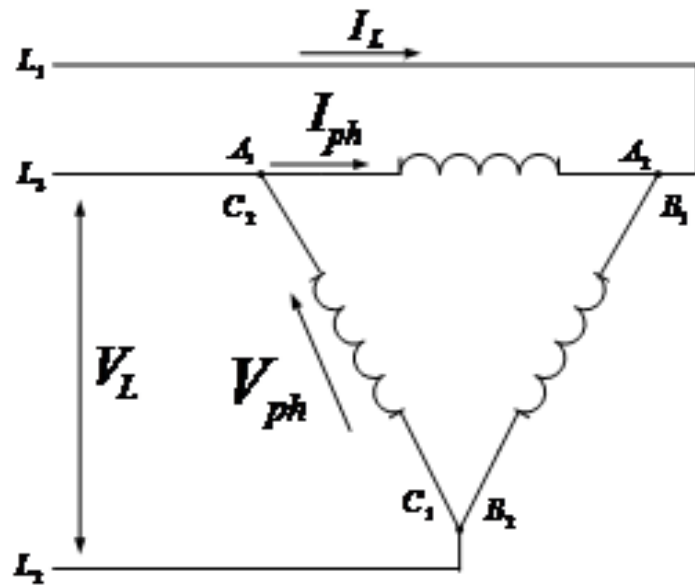


Most common transformer connections

Delta-Star connection

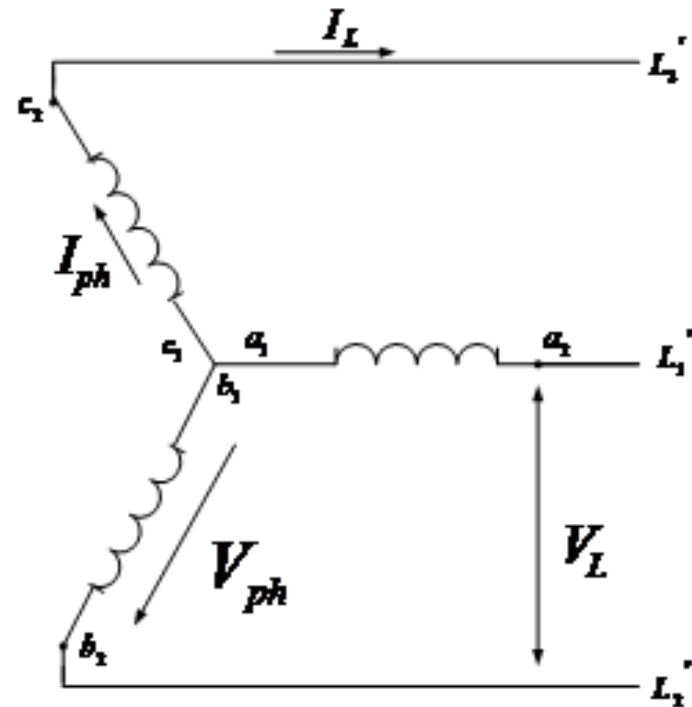
- It is commonly used to step up alternator voltage to transmission line voltage.

- Another common application is in distribution service where as a step-down transformer, the windings are not the most economical.
- The secondary star point can be earthed and a four-wire supply given.



Delta connection

$$V_L = V_{ph} \quad I_L = \sqrt{3} I_{ph}$$



Star connection

$$I_L = I_{ph} \quad V_L = \sqrt{3} V_{ph}$$

Delta-Star connection

Star-Delta connection

- There is no secondary neutral and four-wire supplies cannot be given.
- The main use is as a step-down transformer at the load end of transmission line.

Star-Star

- This is economical for high-voltage transformer.
- However, if the secondary load is unbalanced, the neutral point will be displaced and the line-to-neutral voltages will become unequal, a situation referred to as floating neutral.

- If the neutral of the primary and the neutral of the source are connected together usually by way of ground, the floating neutral problem can be eliminated.

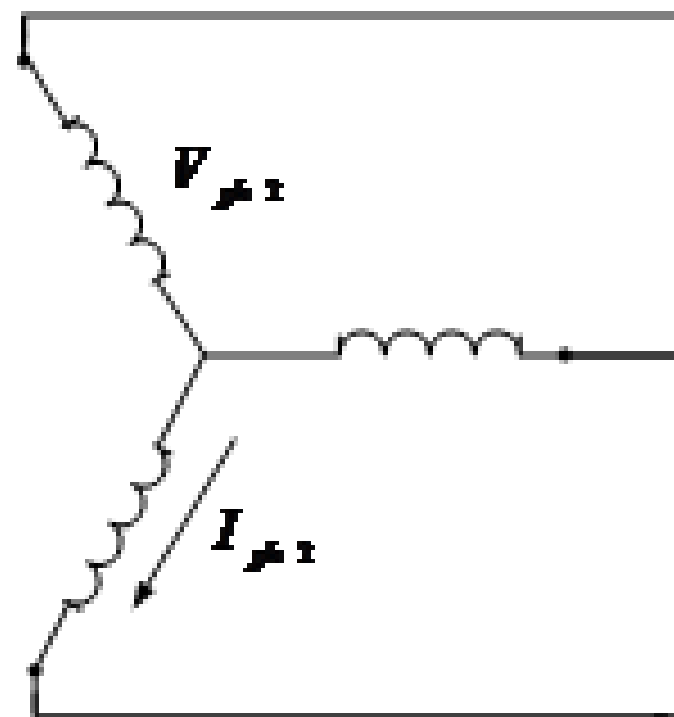
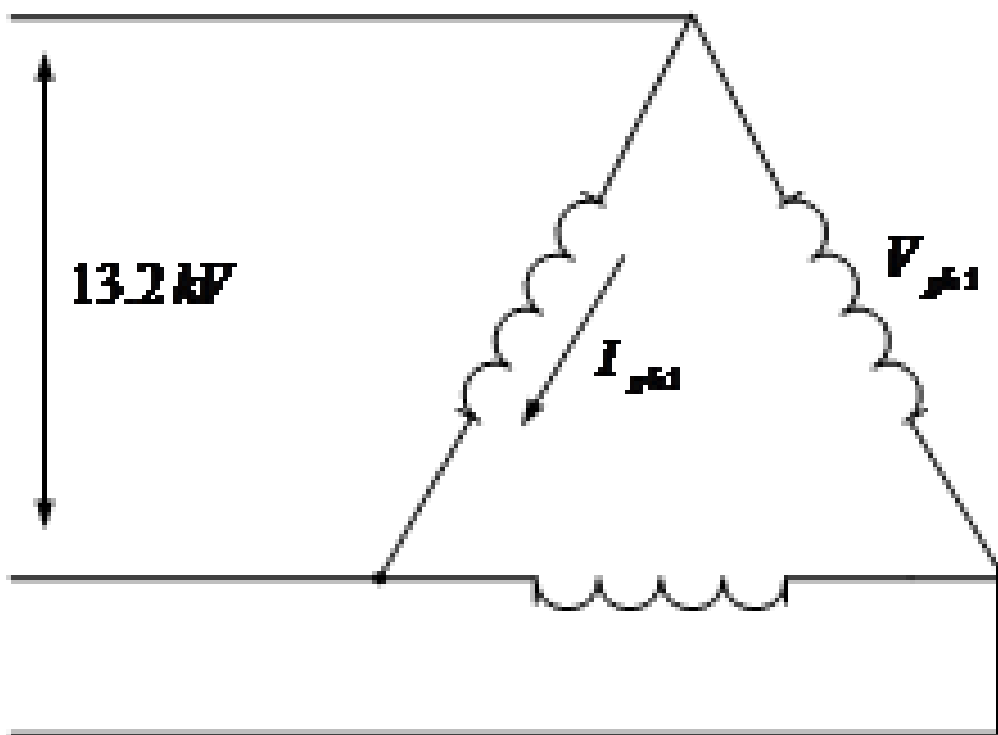
Example 15

Three single-phase step-up transformers rated 40MVA, 13.2 kV / 80 kV are connected in delta-star on a 13.2 kV transmission line. If they feed a 90MVA load, calculate the following:

- (a) the secondary line voltage
- (b) the currents in the transformer windings
- (c) the incoming and outgoing transmission line currents.

Assume transformers are ideal.

Solution



$$(a) \quad \frac{V_{ph2}}{V_{ph1}} = \frac{80}{13.2}$$

$$V_{ph1} = 13.2 \text{ kV (transmission line voltage)}$$

$$\therefore V_{ph2} = 80 \text{ kV}$$

and the line voltage on the secondary side

$$= 80\sqrt{3} = 138 \text{ kV}$$

(b) Current in outgoing line

$$I_{L2} = \frac{90 \times 10^3}{\sqrt{3} \times 80 \sqrt{3}} = 375 \text{ A}$$

Current in the secondary winding

$$I_{ph2} = I_{L2} = 375 \text{ A}$$

(c) Current in incoming line

$$I_{L1} = \frac{90 \times 10^3}{\sqrt{3} \times 13.2} = 3932 \text{ A}$$

Current in the primary winding

$$I_{ph1} = \frac{I_{L1}}{\sqrt{3}} = \frac{3932}{\sqrt{3}} = 2272 \text{ A}$$

Example 16

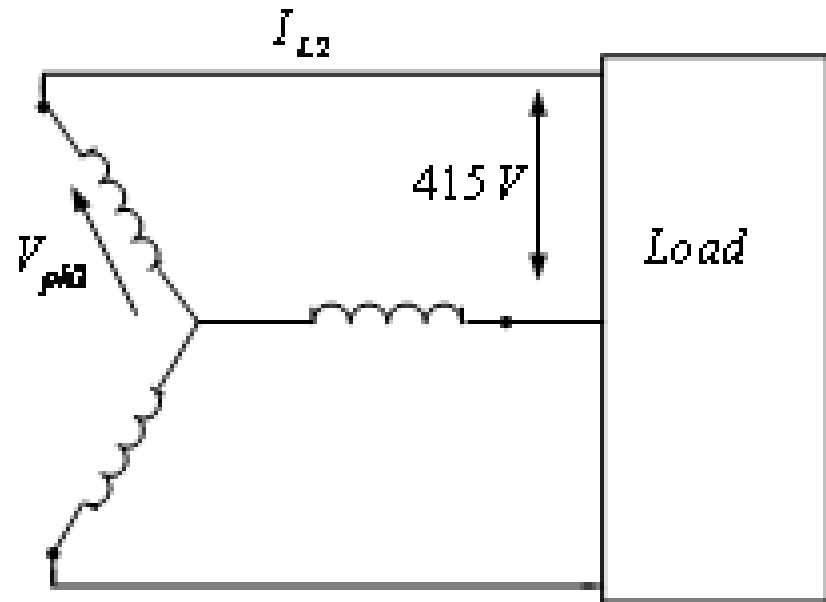
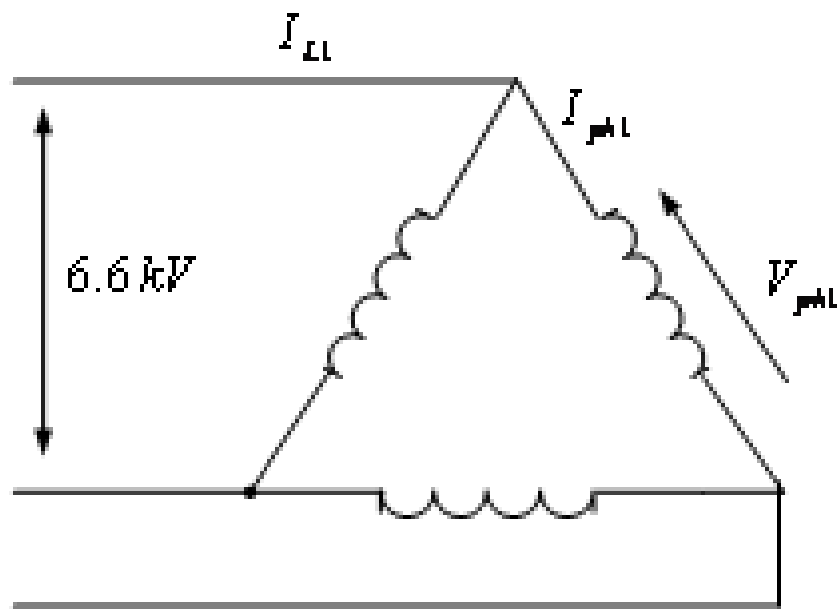
Three single-phase transformers have their primaries joined in delta to a 6600 V, three-phase, three-wire supply. Their secondaries are connected to give a three-phase, four-wire output at 415 V across lines. The total load on the transformers is a balanced load of 150 kW at 0.8 pf lag. The voltage per turn on the primaries is 4.

Calculate:

- (a) the number of turns on the primary winding and the secondary winding
- (b) the currents and voltages in all windings and lines, including the neutral wire on the secondary side
- (c) kVA load on each transformer

Assume transformers are ideal

Solution



150 kW

0.8 pf lagging

(a) Primary turns per phase

$$= \frac{V_{\text{ph1}}}{\text{volt/turn}} = \frac{6600}{4} = 1650 \text{ turns}$$

Secondary turns per phase

$$= \frac{V_{\text{ph2}}}{\text{volt/turn}} = \frac{415}{\sqrt{3} \times 4} = 60 \text{ turns}$$

$$(b) \quad \text{Secondary } I_{L2} = I_{ph2} = \frac{\text{Power}}{\sqrt{3}V_L \cos\phi}$$

$$= \frac{150 \times 10^3}{\sqrt{3} \times 415 \times 0.8} = 261 \text{ A}$$

$I_N = 0 \text{ A}$ because load is balanced

$$I_{ph1} = \left(\frac{N_2 I_{ph2}}{N_1} \right) = \frac{60 \times 261}{1650} = 9.5 \text{ A}$$

$$I_L = 9.5\sqrt{3} = 16.4 \text{ A}.$$

Autotransformers

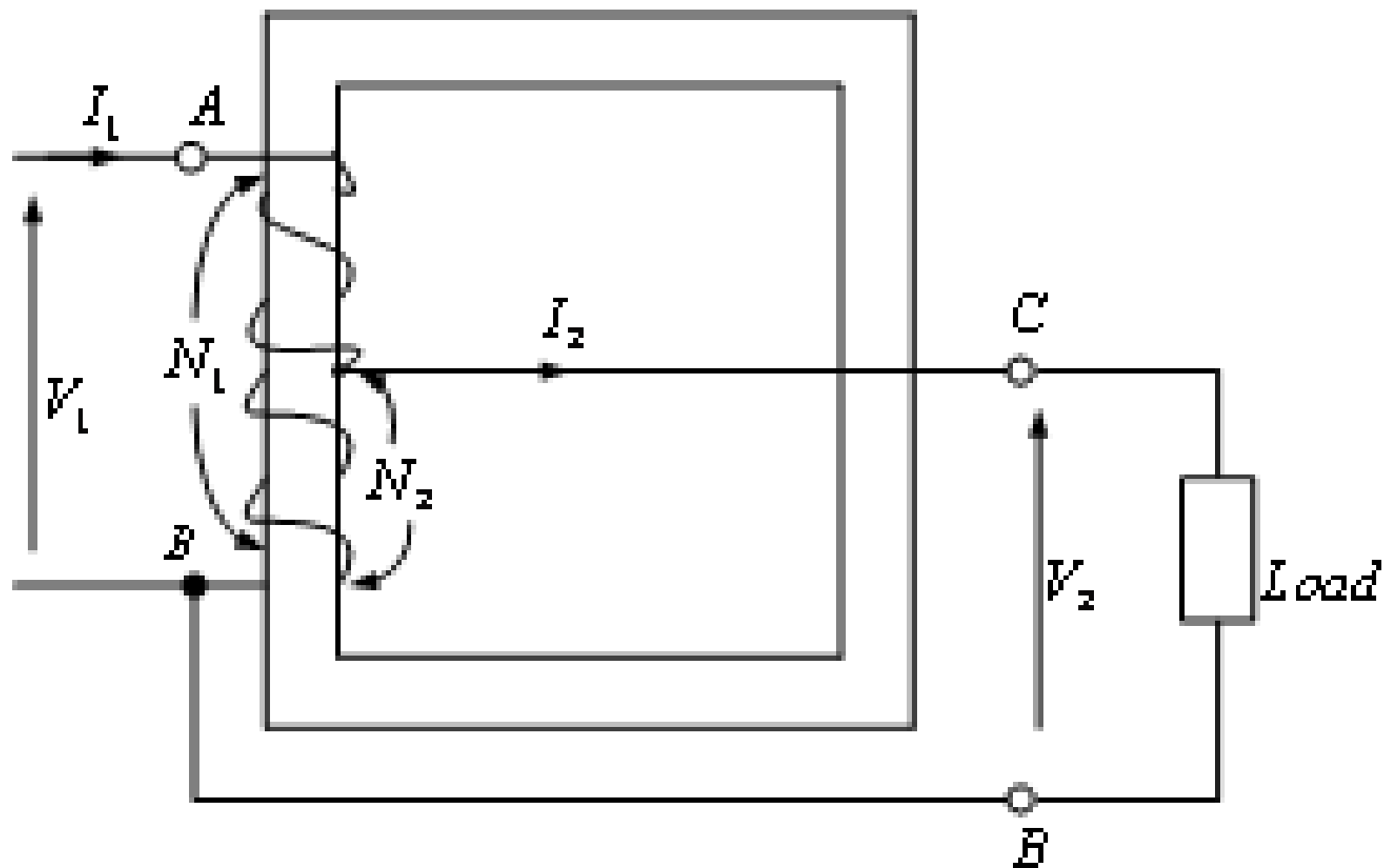
A decorative graphic element consisting of a large, light blue arc that starts from the top left and curves towards the bottom right, creating a sense of motion or a spotlight effect on the text.

- An autotransformer has a single tapped winding which serves both

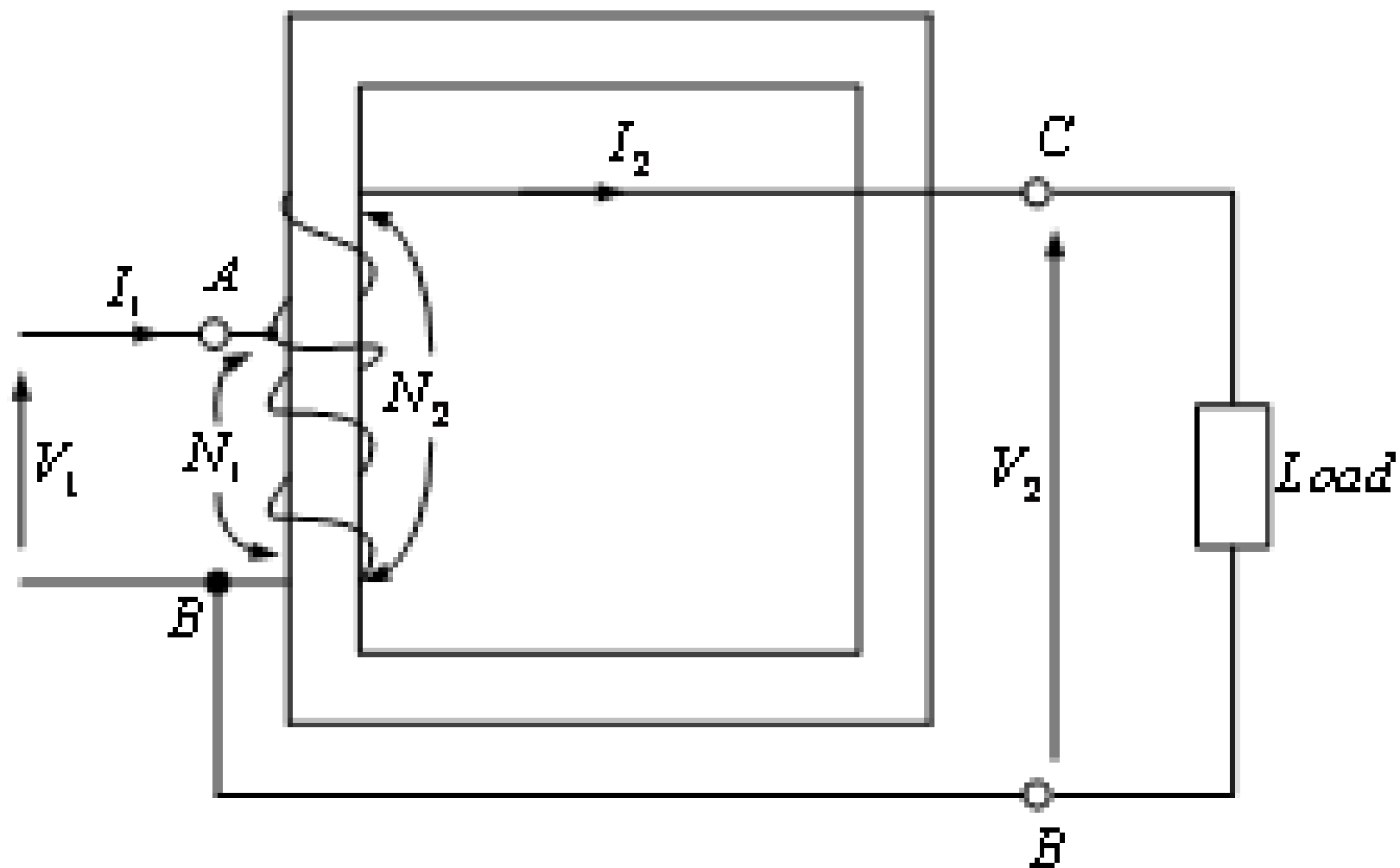
- primary and

- secondary functions

as shown below.



Step-down autotransformer



Step-up autotransformer

Autotransformer equations

- If an autotransformer is considered to be ideal transformer, i.e. we neglect losses, leakage flux and magnetizing current then

$$\frac{V_2}{V_1} = \frac{N_2}{N_1}$$

$$N_1 I_1 = N_2 I_2$$

$$V_1 I_1 = V_2 I_2$$

Advantage(s) autotransformer over two-winding transformer

- The main advantage gained in the use of autotransformer is **the saving of copper**.
- For a two-winding transformer and an autotransformer which can perform the same duty it can be shown that

$$\frac{\text{volume of copper in autotransformer}(G_a)}{\text{volume of copper in two - winding transformer}(G_2)} = 1 - \frac{V_L}{V_H}$$

Or

Saving of copper made by using autotransformer

$$G_2 - G_a = \frac{V_L}{V_H} \times G_2$$

Where

V_L = voltage on the low-voltage side

V_H = voltage on the high-voltage side

- In practice, voltage ratios less than about $1/3$ show little economic benefit over two-winding transformer because of other factors such as cost of insulation.

Disadvantage(s) of autotransformer over two-winding transformer

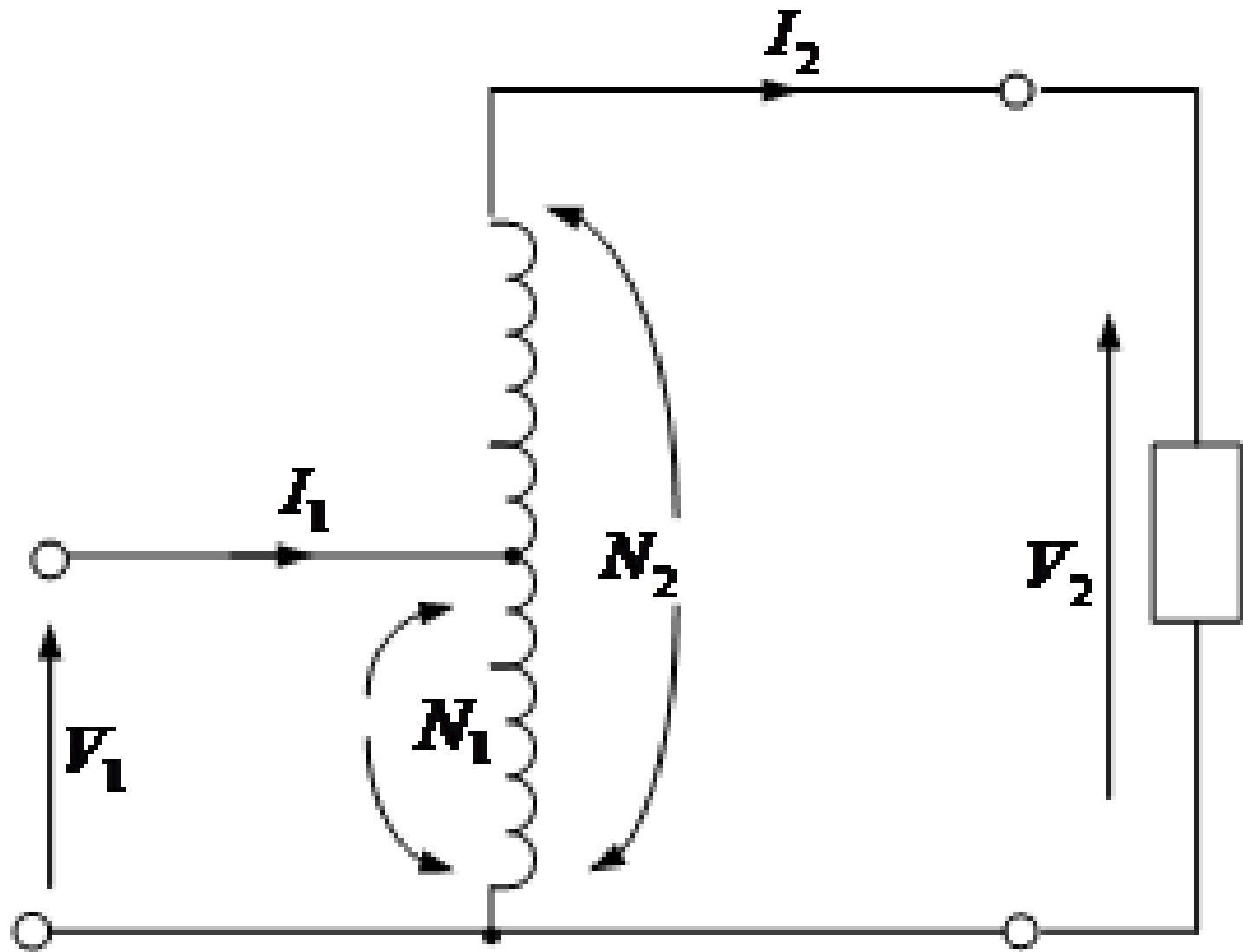
- The main disadvantage is that the primary and secondary circuits are not isolated from each other.

Example 17

An autotransformer is required to step up a voltage from 220 to 250 V. The total number of turns is 2000. Determine

- (a) the position of the tapping point
- (b) the approximate value of the current in each part of the winding when the output is 10 kVA and
- (c) the economy in copper over the two winding transformer which will perform the same duty.

Solution



$$(a) \quad \frac{V_1}{V_2} = \frac{N_1}{N_2} = \frac{220}{250}$$

$$N_1 = \frac{220}{250} \times N_2 = \frac{220}{250} \times 2000$$

$$= 1760$$

Position is $2000 - 1760 = 240$ turns
from one end.

$$(b) \quad V_2 I_2 = 10 \times 10^3$$



$$I_2 = \frac{10 \times 10^3}{250} = 40 \text{ A}$$

$$V_1 I_1 = 10 \times 10^3$$



$$I_1 = \frac{10 \times 10^3}{220} = 45.45 \text{ A}$$

Therefore current in series winding is **40 A** and current in common winding = **45.45 – 40 = 5.45 A**

$$(c) \quad \text{Saving in copper} = \frac{V_L}{V_H} = \frac{220}{250} = 0.88 \text{ pu or } 88\%$$

of copper used in the two-winding transformer

Two-winding transformer connected as an autotransformer

- A two-winding transformer can be changed into an autotransformer by connecting the primary and secondary windings in series.

- The following rules apply whenever a two-winding transformer is connected as autotransformer:

1. The current in any winding should not exceed its current rating
2. The voltage across any winding should not exceed its voltage rating

3. Rated current in one winding gives rise to rated current in the other
4. Rated voltage across one winding gives rise to rated voltage across the other
5. If current in one winding flows from say A_2 to A_1 , then current in the other winding must flow from a_1 to a_2 and vice versa

6. The voltages add when terminals of opposite polarity (A_1 and a_2 or A_2 and a_1) are connected together by a jumper.
7. The voltages subtract when A_1 and a_1 (or A_2 and a_2) are connected together.

Example 18

A two-winding single-phase transformer rated 15 kVA, 600 V / 120 V, 60 Hz.

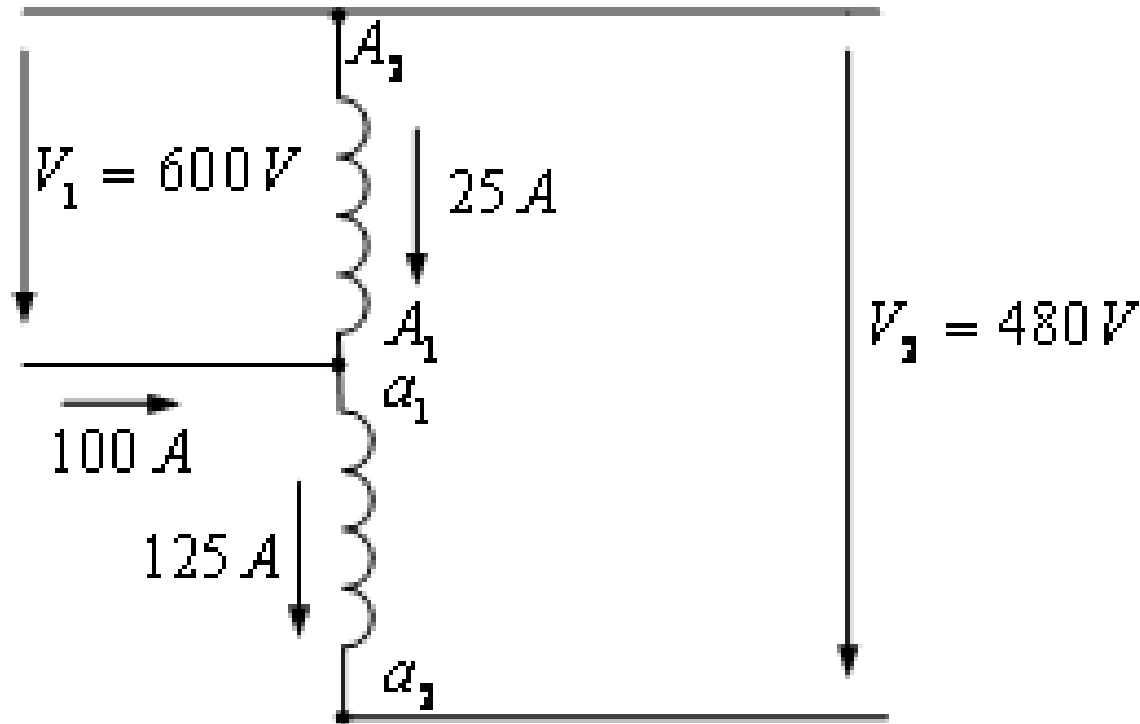
We wish to reconnect it as an autotransformer in three different ways to obtain three different voltage ratios:

- (a) 600 V primary to 480 V secondary
- (b) 600 V primary to 720 V secondary
- (c) 120 V primary to 480 V secondary

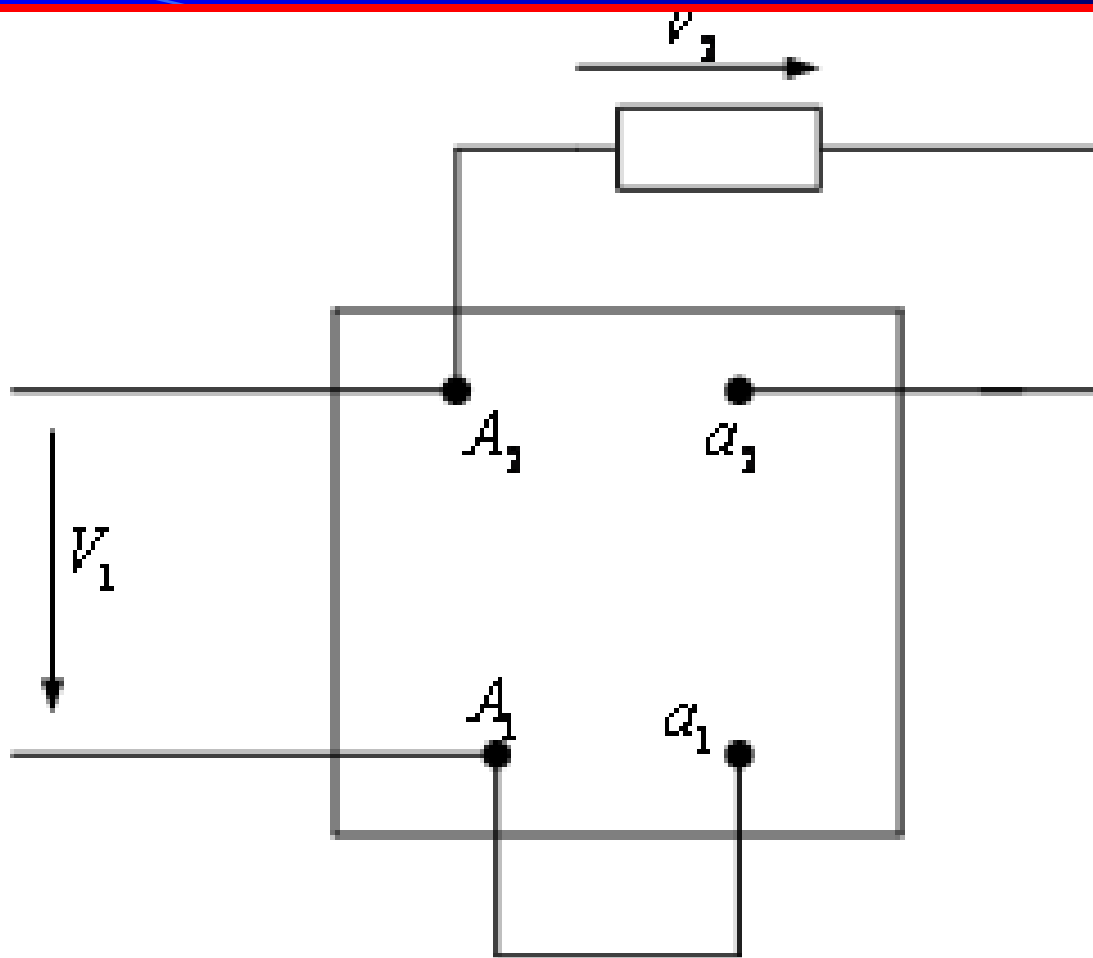
Calculate the maximum load the transformer can carry in each case.

Solution

- (a) The secondary voltage 120 V must be subtracted from the primary voltage to obtain the 480 V.

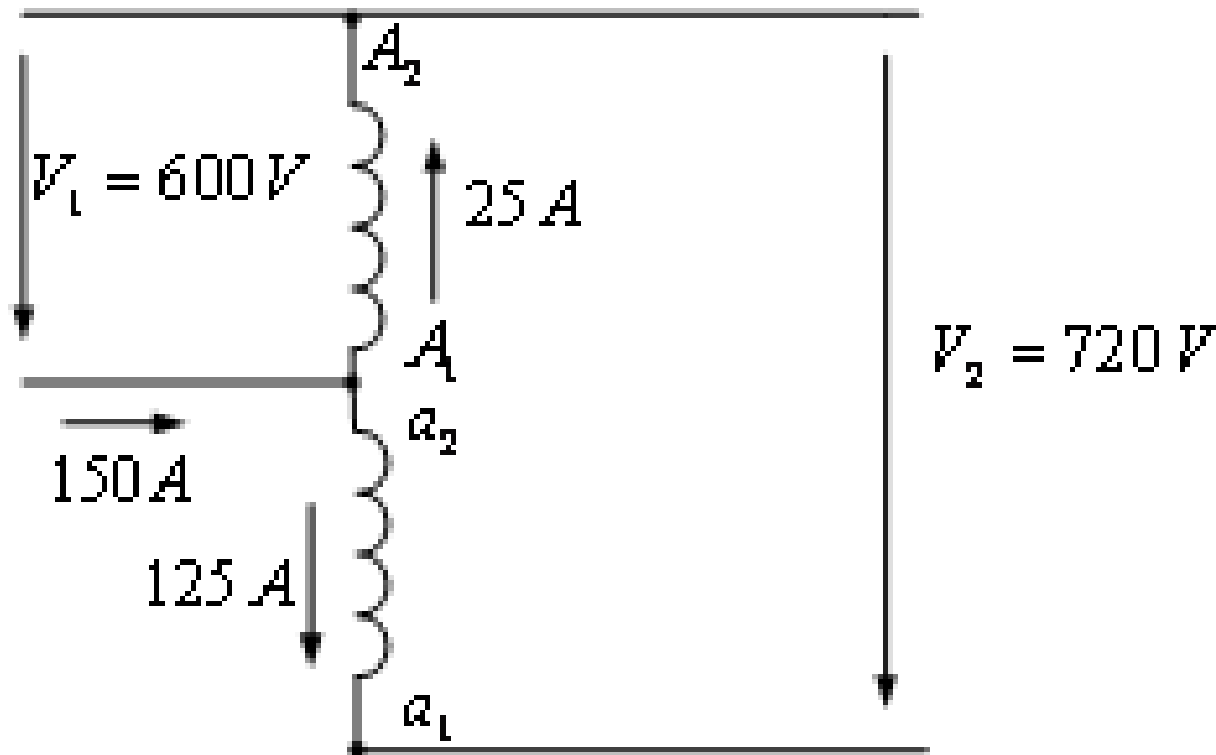


$$kVA = 600 \times 100 \times 10^{-3} = 60$$

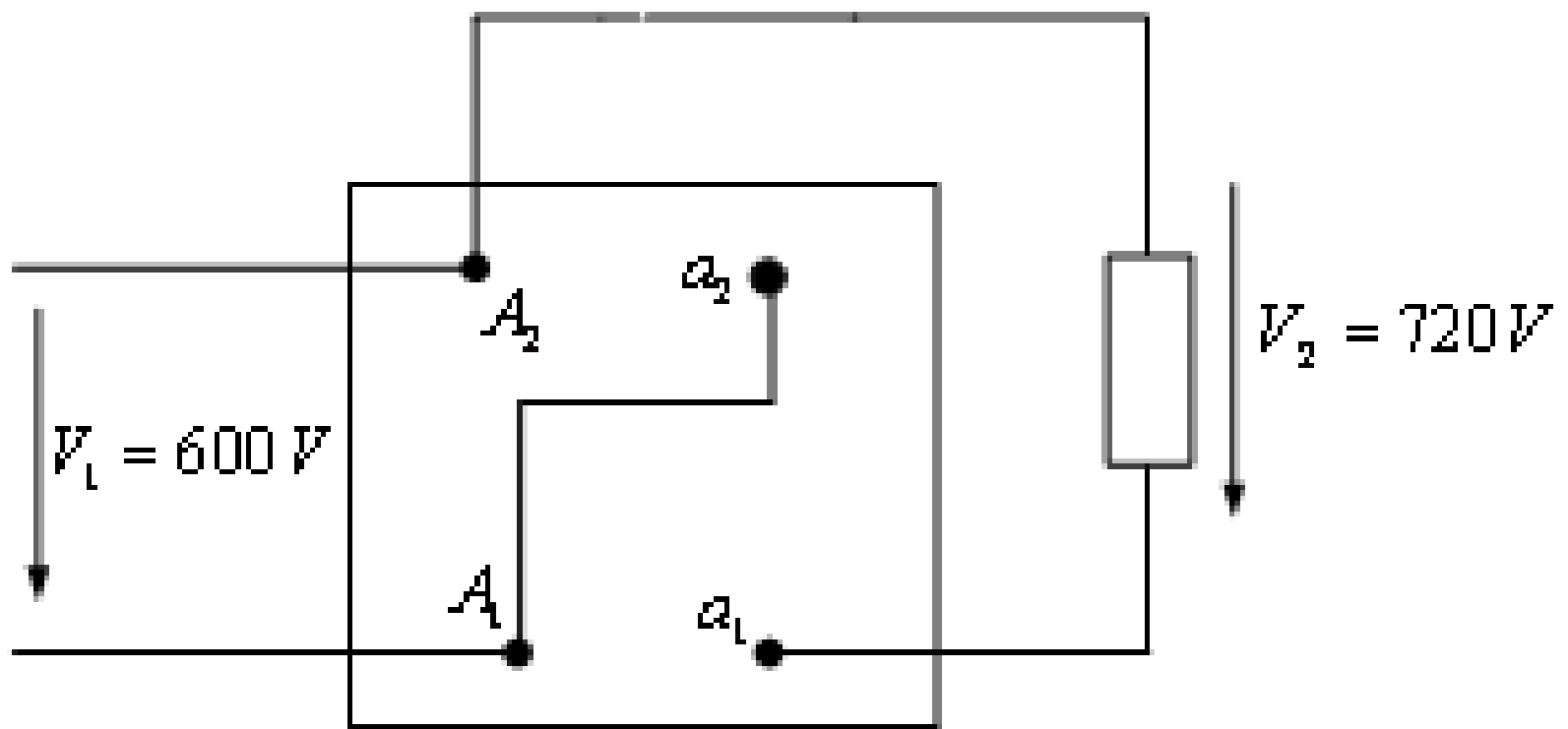


Conection diagram

(b) The secondary voltage 120 V must be added to the 600 V to obtain 720 V

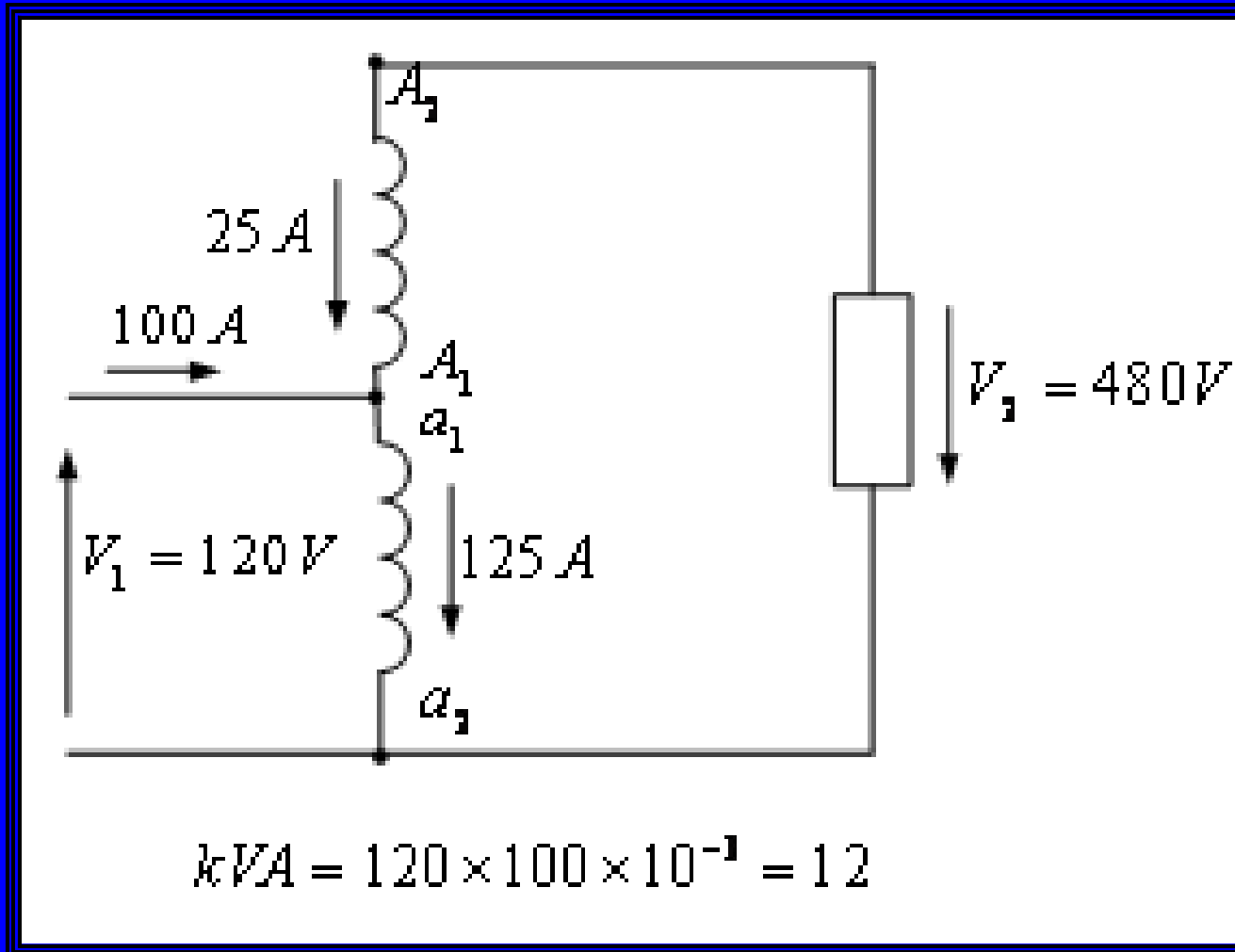


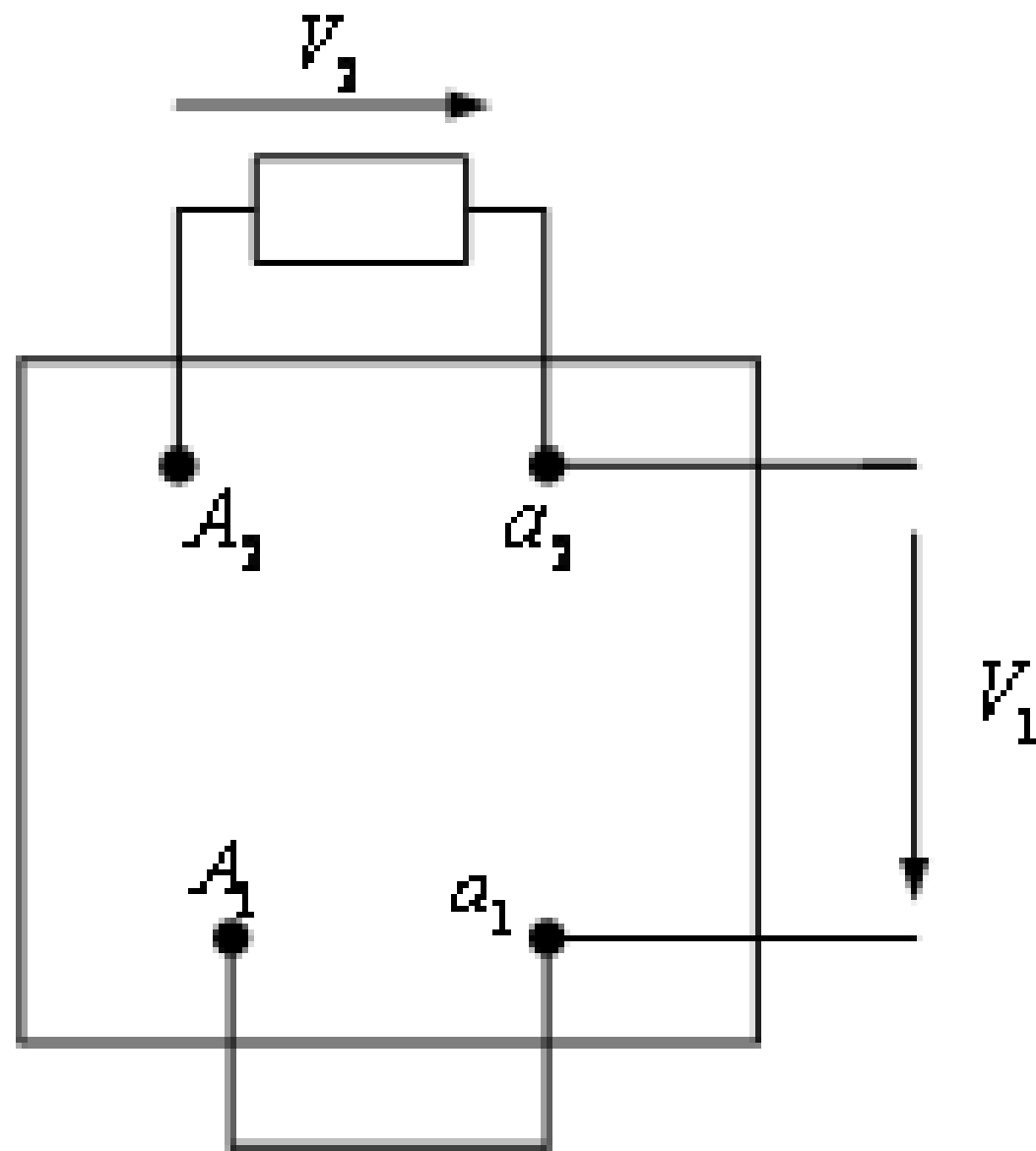
$$\text{kVA} = 600 \times 150 \times 10^{-3} = 90$$



Conection diagram

(c) The 120 V becomes the primary of the autotransformer and the 120 V is subtracted from the 600 V to obtain its secondary





Conection diagram

Applications of autotransformers

- They are mainly used for
 - Variac
 - interconnecting power systems that are operating at roughly the same voltage (e.g. 132 kV, 275 kV, 400 kV) and
 - starting squirrel-cage induction motors.

Instrument transformers

A decorative graphic element consisting of a large, light blue arc that starts from the top left and curves towards the bottom right, creating a sense of motion or a spotlight effect on the text.

- They are used in ac circuits to serve these purposes:

- to make possible the measurement of high voltages with low-voltage instruments or large currents with low current ammeters

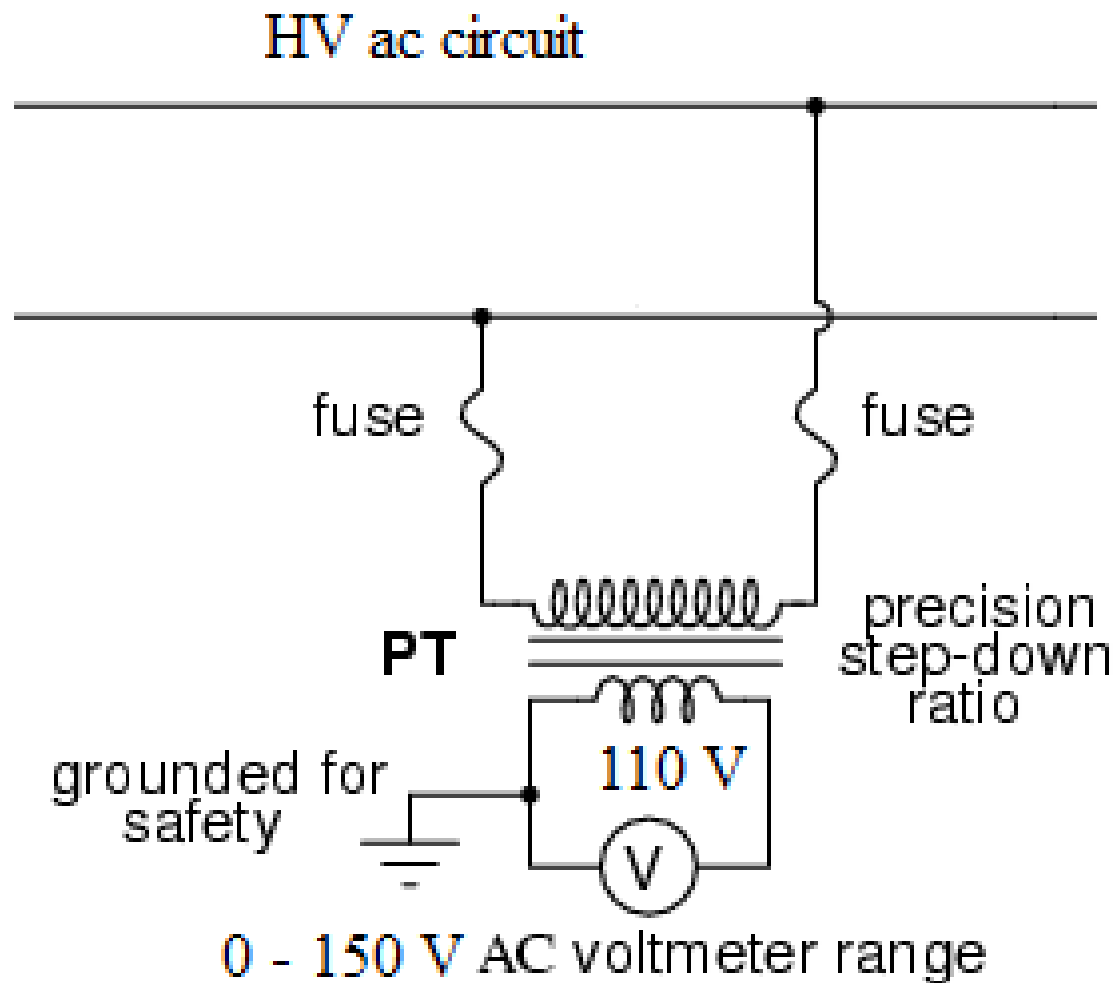
- to insulate high voltage circuits being monitored from measuring circuit in order to protect the measuring apparatus and operator
- to energize relays for the operation of protective and automatic control devices

- The load on the secondary of an instrument transformer is called **its burden** and is expressed in **volt-amperes**
- There are two types of instrument transformers:
 - the **voltage (or potential) transformer** and
 - the **current transformer**

The voltage or potential transformers (VTs or PTs)

- The construction is similar to a power transformer.
- The primary is connected directly to the power circuit either between **two phases** or between **a phase and ground** and the secondary is connected to **instruments** and **coils of relays**.

- Voltage transformers are designed to step down the primary voltage to a nominal or rated voltage of 110 V so that standard instruments and relays can be used.
- The circuit for a potential transformer is shown below.



Potential transformer installed on HV circuit

- One terminal of the secondary winding is always earthed for the following reasons:

- The windings though insulated from each other, are connected invisibly together by distributed capacitance between them.

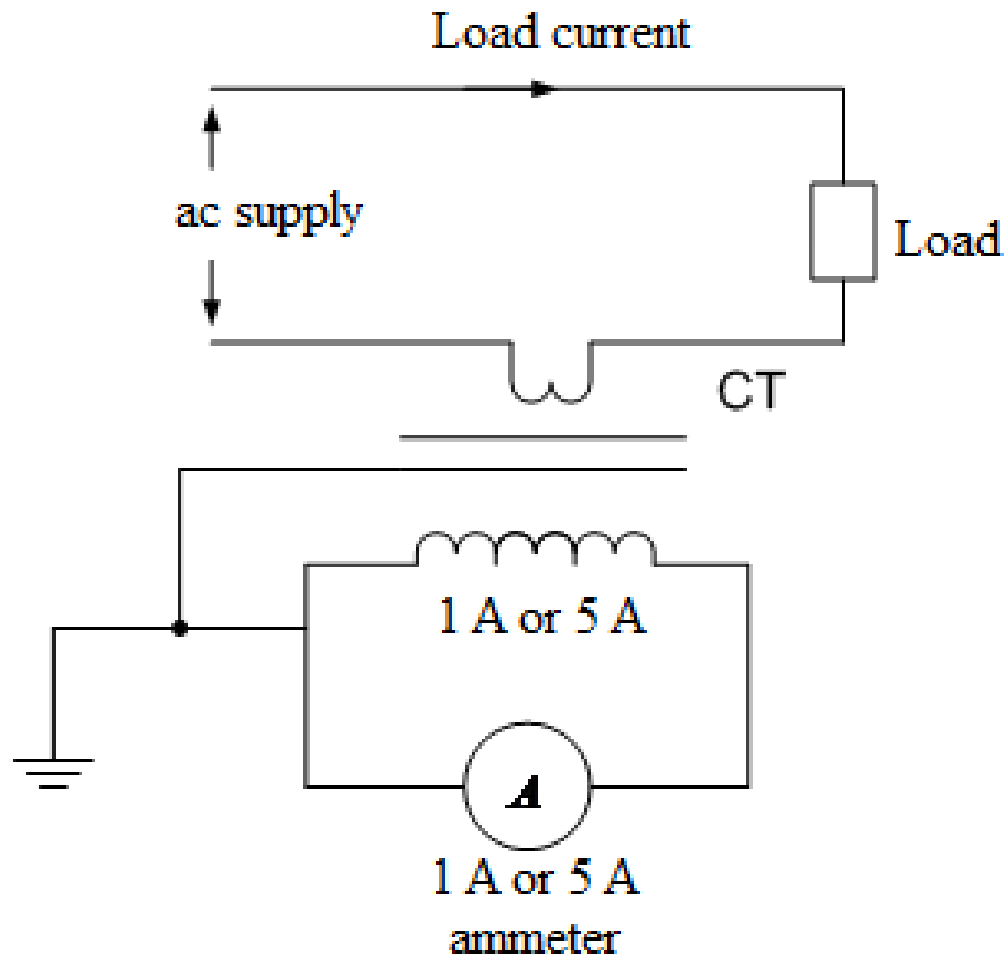
- By earthing one of the secondary terminals, the highest voltage between the secondary lines and earth can never rise above that of the secondary voltage.

Current transformers (CTs)

- Their primary consists of small number of turns connected in series with the power circuit load.
- The secondary consists of a larger number of turns and it is connected to ammeter, current coils of other instruments or current coils of relays.

- Current transformers have the ratio of primary to secondary current approximately constant.
- The nominal or rated secondary currents are usually 5 A or 1 A, irrespective of the primary current rating.

- The transformer is connected in the power circuit as shown below.
- As in the case of PT (and for the same reasons) one of the secondary terminals is always earthed.



Current transformer installed in HV circuit

- Current transformer secondary circuit must not be opened while current is flowing in the primary for the reasons given below:

- The iron core becomes saturated and very high voltage spikes (several thousand volts) are induced across the open-circuited secondary.

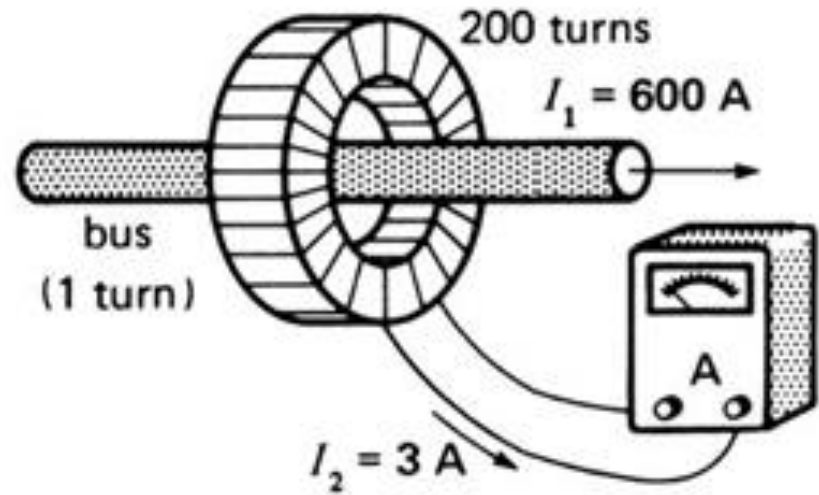
- These voltages are dangerous to life and to the transformer insulation.
- The core when it becomes saturated can also cause excessive heating of the core and windings.

- When it becomes necessary to remove a load from the secondary circuit, the secondary winding must first be short circuited.
- When the line current exceeds 100 A we can sometimes use a toroidal or bar-primary ($N_1=1$) transformer.

- It consists of a laminated ring-shaped core which carries the secondary winding.
- The primary is composed of a single conductor that simply passes through the centre of the ring as shown below.



Toroidal transformer



1000 A/5 A toroidal transformer
connected to measure current in a line

- Toroidal CTs are **simple** and **inexpensive**.
- They are widely used in HV and MV indoor installations.



CTs and PTs at 115 kV switchyard

- Current transformers are also commonly used for the measurements of large currents even when the circuit voltage is not dangerously high.
- This avoids bringing heavy leads to the instrument panels.

Example 19

A potential transformer rated $14400\text{ V} / 115\text{ V}$ and a current transformer rated $75\text{ A} / 5\text{ A}$ are used to measure the voltage and current in a transmission line. If the voltmeter indicates 111 V and the ammeter reads 3 A , calculate the voltage and current in the line.

Solution

The voltage on the line is

$$V = 111 \times \frac{14400}{115} = 13900 \text{ V}$$

The current in the line is

$$I = 3 \times \frac{75}{5} = 45 \text{ A}$$

Example 20

The toroidal current transformer shown above has a ratio of $1000 \text{ A} / 5 \text{ A}$. The line conductor carries a current of 600 A .

- (a) Calculate the voltage across the secondary winding if the ammeter has an impedance of 0.15Ω
- (b) Calculate the voltage drop the transformer produces on the line conductor.

(a) Current in the secondary

$$I_2 = \frac{5}{1000} \times 600 = 3 \text{ A}$$

Voltage drop across the burden

$$= 3 \times 0.15 = 0.45 \text{ V}$$

$$(b) \quad \frac{V_1}{V_2} = \frac{N_1}{N_2} = \frac{I_2}{I_1}$$

$$V_1 = \frac{I_2}{I_1} \times V_2 = \frac{5}{1000} \times 0.45 = 2.25 \text{ mV}$$